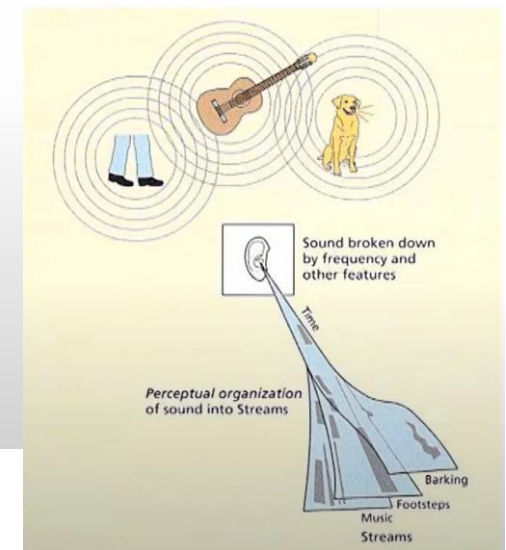


計算認知神經科學 腦與感官



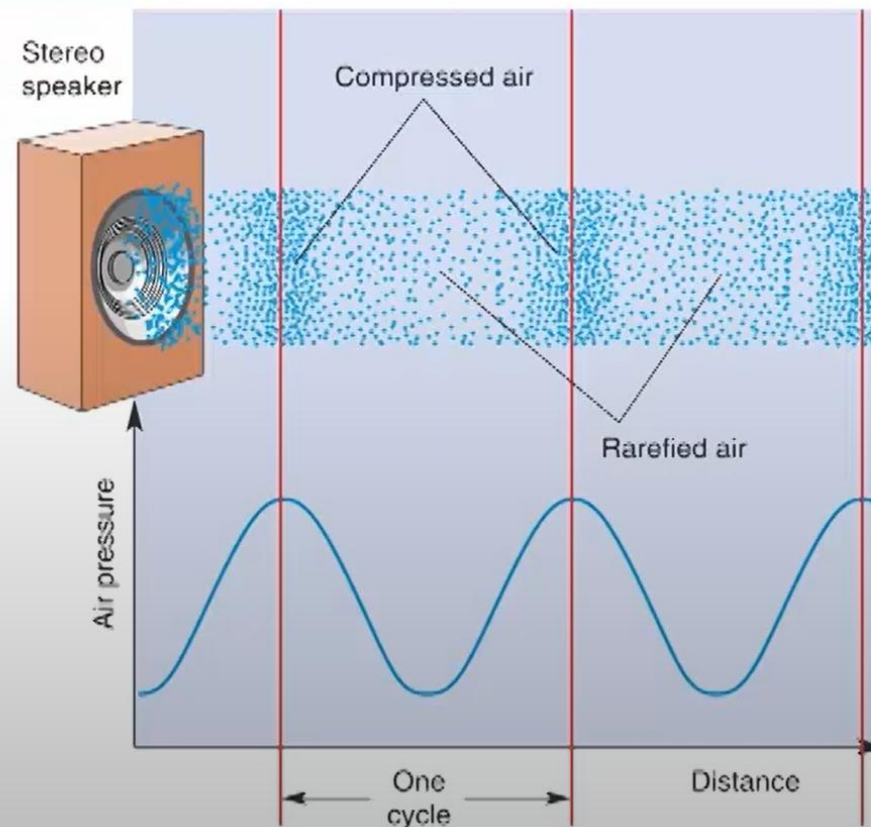
Introduction

- The hearing brain is concerned with extracting “constancy” out of an infinitely varying array of sensory input and it will actively interpret the sensory input.
 - Recognize a familiar tune when presented in different key
 - Recognize a familiar voice on a wide range of acoustic environment
- Sense of hearing, audition
 - Detect sound
 - Perceive and interpret nuances



The nature of sound

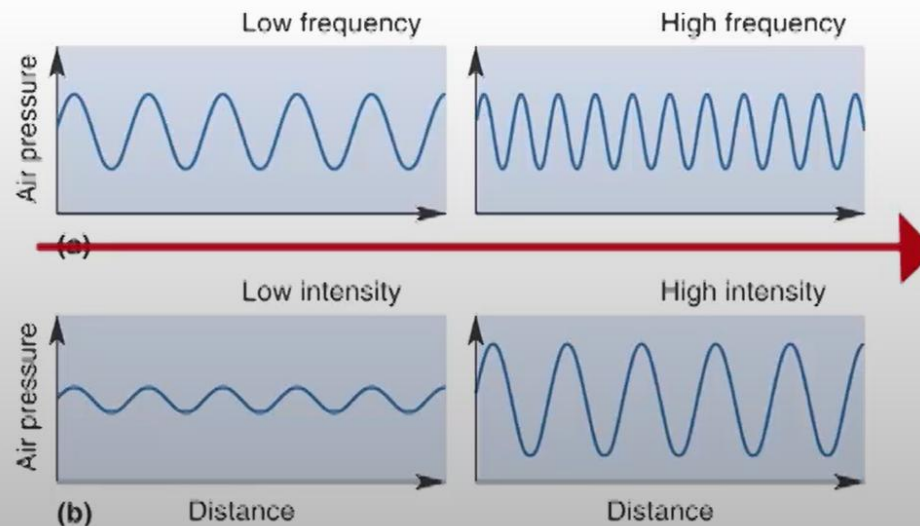
- Sound
 - A form of mechanical energy
 - a vibrating object generates pressure waves in air, water, or some other medium



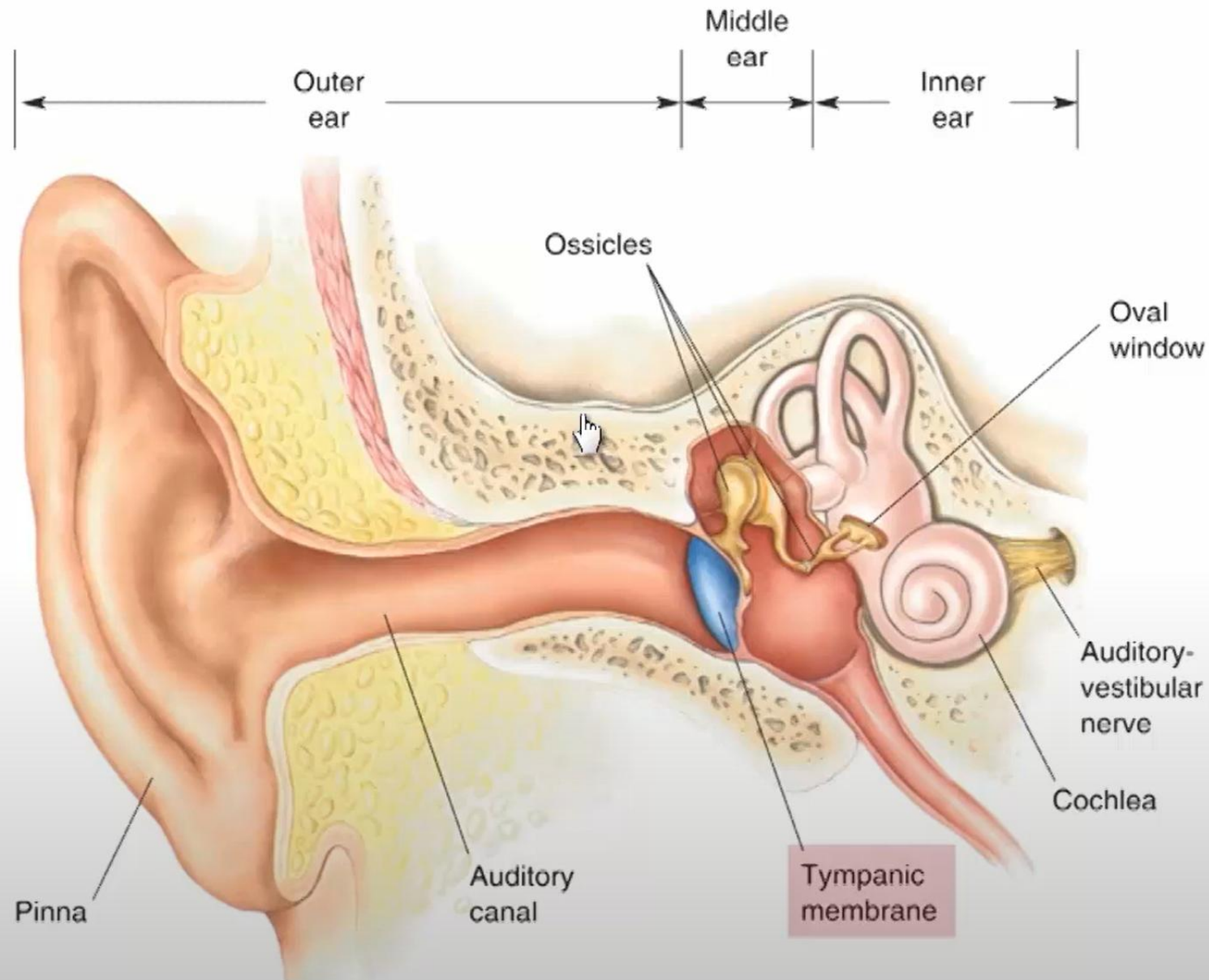
The nature of sound

- Sound

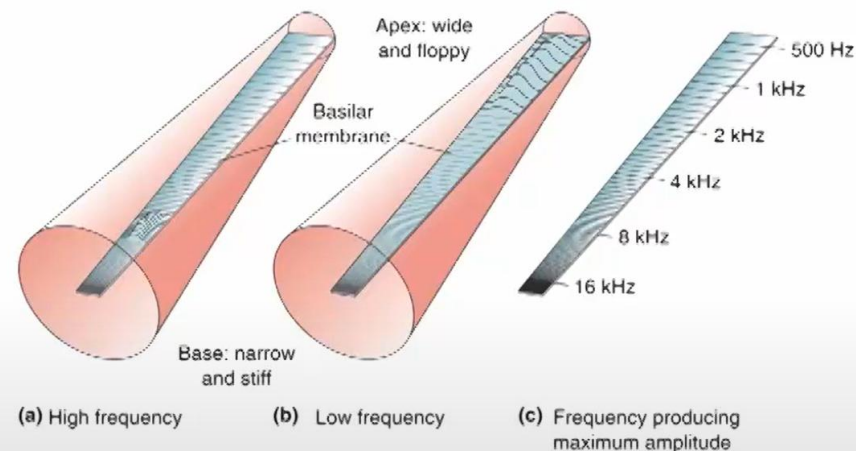
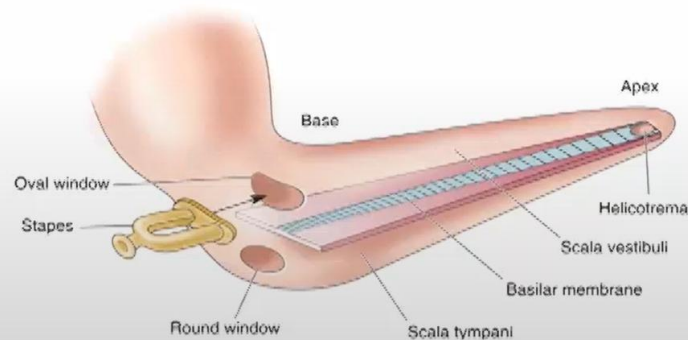
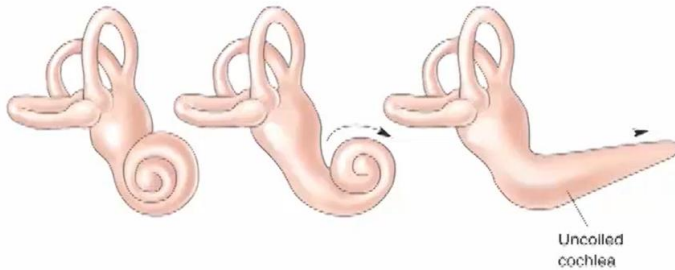
- **Frequency**: Number of waves per second, **Hertz, Hz**
 - ✓ Range of human auditory system: 20 Hz to 20,000 Hz
 - ✓ We perceive high-frequency waves as having a higher pitch
- **Intensity (amplitude)**: Difference in pressure between compressed and rarefied patches of air
 - ✓ We perceive high-intensity waves as louder



The structure of the auditory system

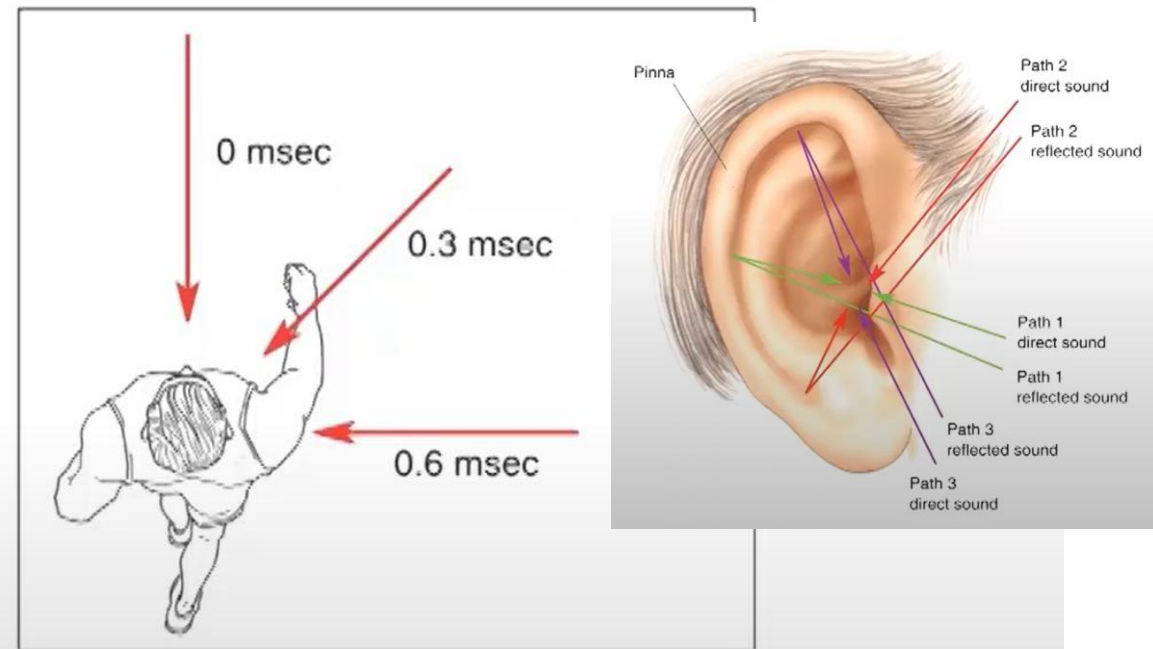
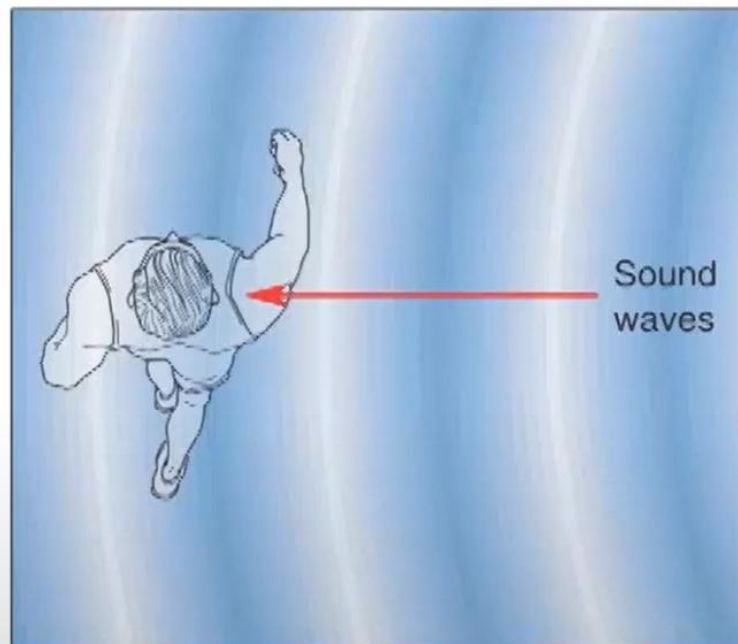


- Auditory pathway stages: **from eardrum to cochlea**
 - Transform sound pressure into neural signals
 - Sound wave → eardrum vibrates → vibrations produce tiny wave in the inner ear's fluid → stimulate hair cells (located in cochlea) → hair cells generate action potentials



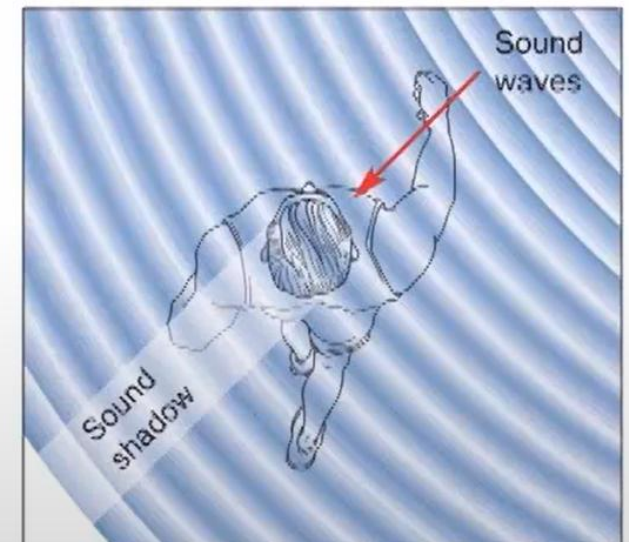
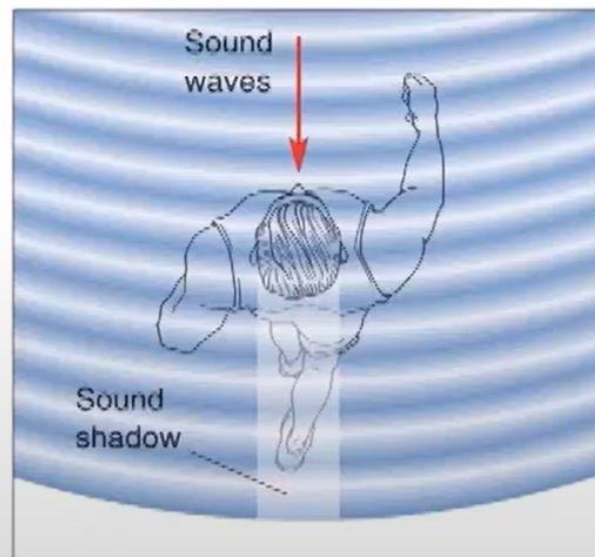
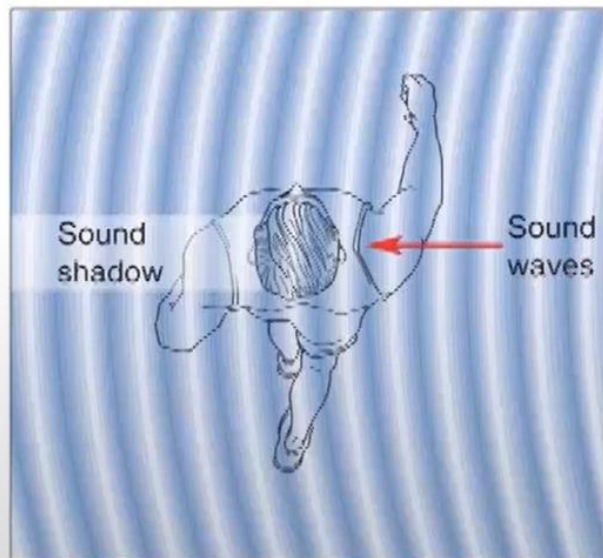
Mechanisms of sound localization

- Two cues to localize sound in horizontal plane
 - Interaural **time delay**: the differences in when a sound reaches each of the two ears



Mechanisms of sound localization

- Two cues to localize sound in horizontal plane
 - Interaural **time delay**: the differences in when a sound reaches each of the two ears
 - Interaural **intensity difference**: The differences in the sound's intensity at the two ears



Music perception

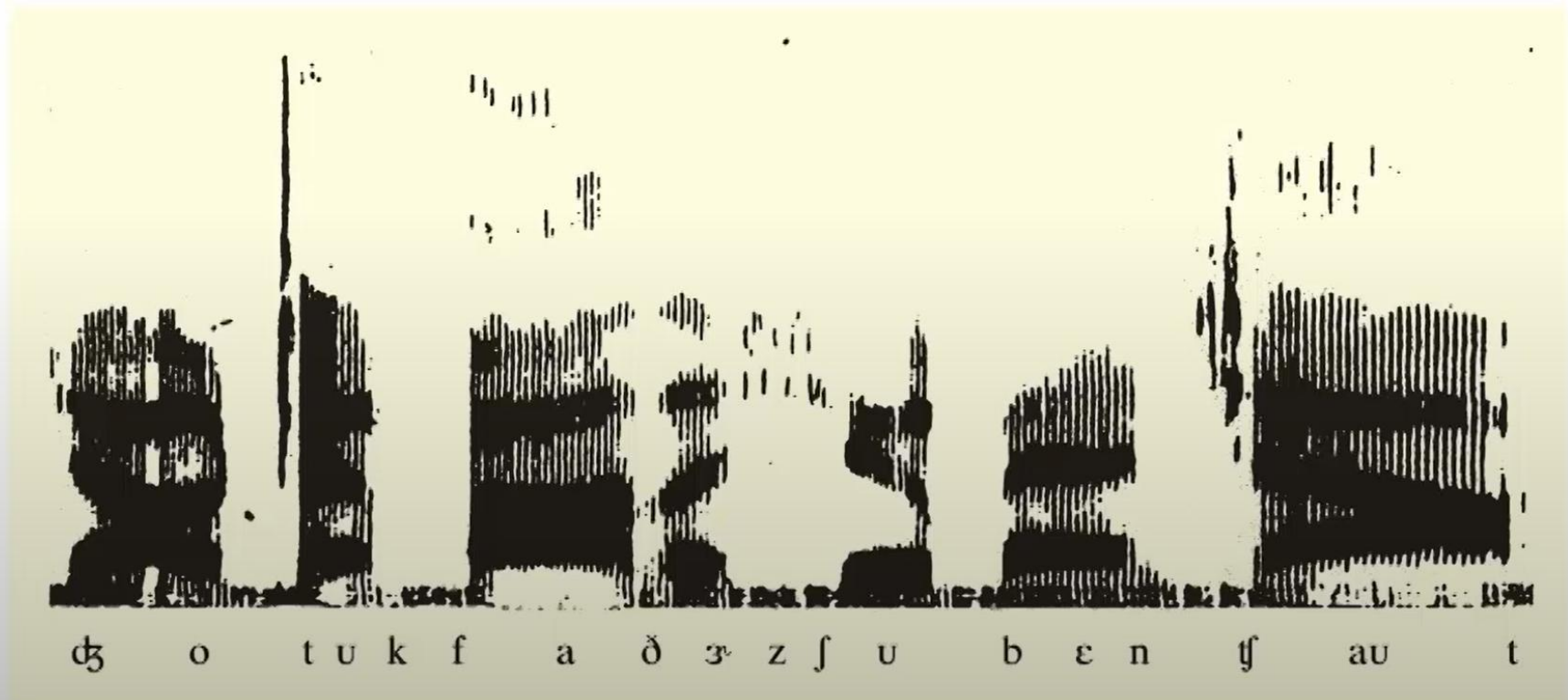
- Music considered a “human instinct” by some (universal, emerges without training)
- But evolutionary function of music remains uncertain
- Musical systems use a set of discrete pitches presented in patterns that vary over both **pitch** (e.g. melody) and **time** (rhythm, tempo)
- Music, like vision or language, may be decomposed into different mechanisms



Speech perception

- Acoustic signal

- Spectrogram shows frequency against time, intensity shown by darkness
- e.g. “Joe took father’s shoe bench out”

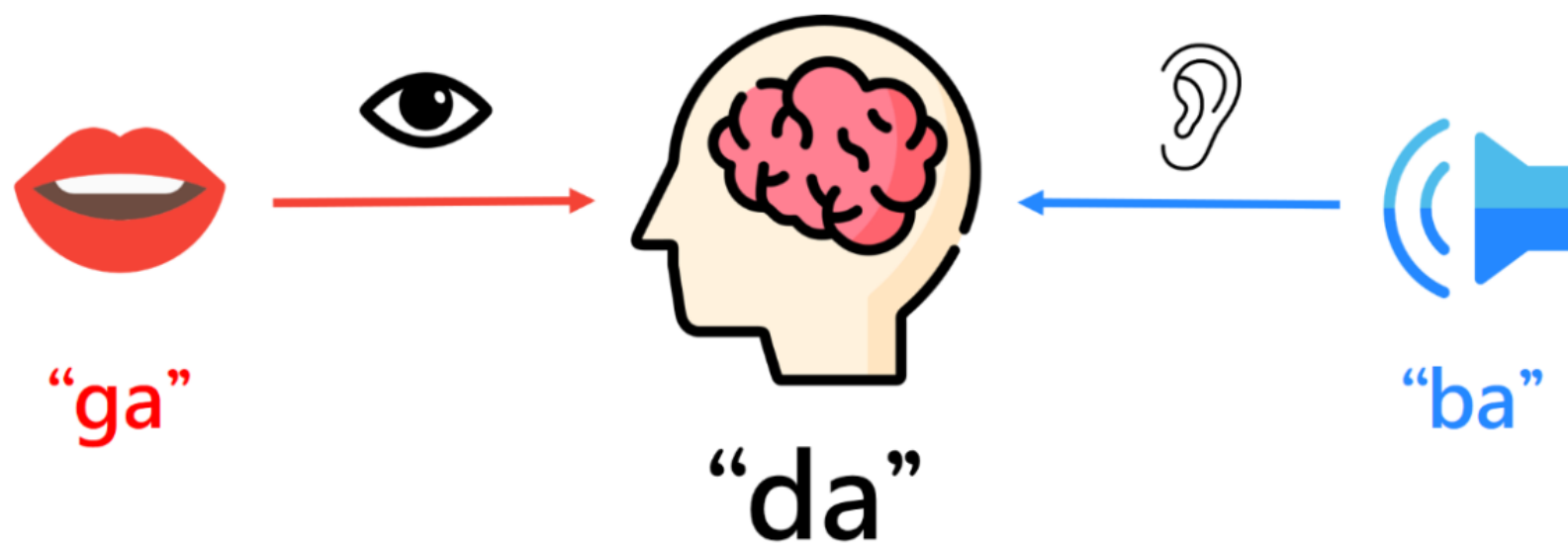


Speech perception

- Gaps don't occur between words but occur with certain consonants (that restrict flow of air)
- Problem of segmenting continuous input
 - e.g., “ice-cream” vs. “I scream”
- Problem of inter-speaker differences;
 - Pitch affected by age and sex; different dialects, talking speeds etc.

McGurk effect illusion

- An auditory percept derived from a fusion of mismatching heard speech and seen speech
- A multisensory Speech Illusion



說話者唇形為「ga」、播放聲音為「ba」，會聽到「da」的聲音。圖／雅文基金會

Neural basis of McGurk effect illusion

- **left superior temporal sulcus (STS)** is critical for audiovisual integration during speech perception

A. Congruent



Percept: "ba"

B. Non-McGurk Incongruent

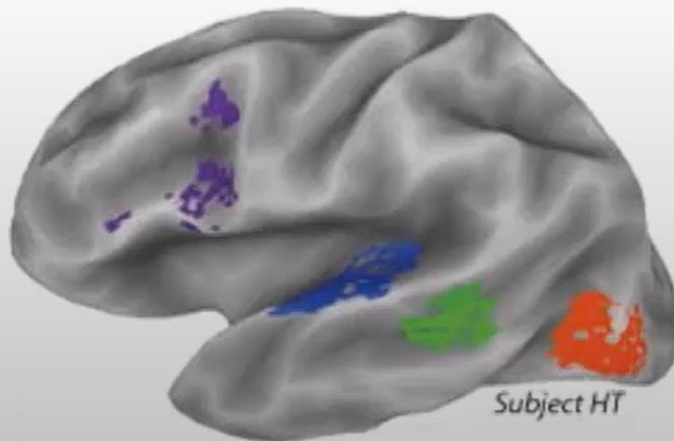


Percept: "ga"

C. McGurk Incongruent



"da" in perceivers
"ba" in non-perceivers



Inferior prefrontal gyrus

Auditory cortex

Left superior temporal sulcus (STS)

Visual cortex

As a speaker

- When we speak we usually wish to convey some **message**
 - Decide what to say
 - Select words with appropriate meanings
 - Organize words into grammatical form
 - Turn sentences into speech
 - Sometimes the message gets garbled. We may stammer, or pause.

As a listener

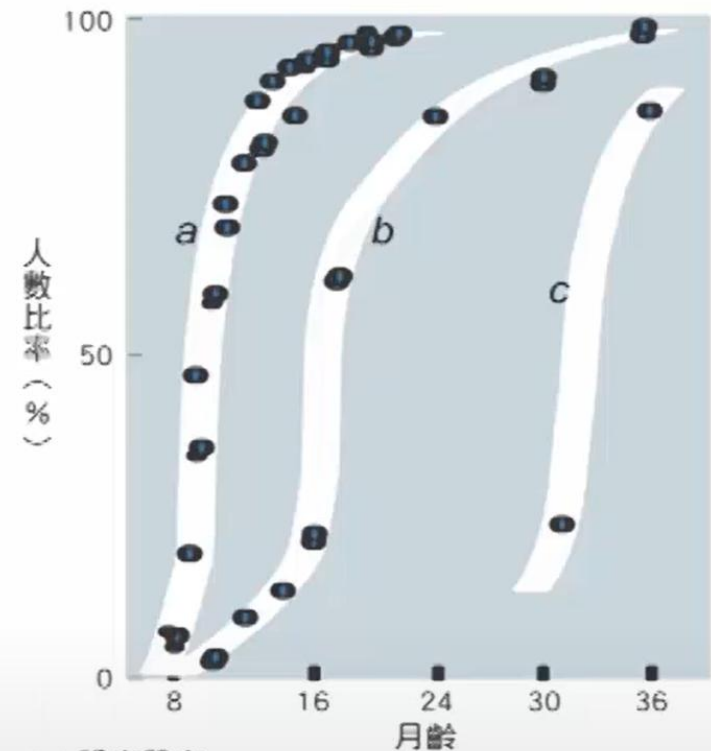
- Trying to understand a speech based on his/her thought or knowledge

Can non-human animals have language?



肯茲是一隻巴諾布猿，飼養在美國喬治亞州立大學一個使用語言的環境中。透過指點代表特定文字的符號，牠可以建構出像人類兩歲大孩童使用的需求語句。肯茲的語言理解力相當於兩歲半的兒童。巴諾布猿的語言實驗，讓我們了解到人類語法是多麼獨特。

兒童語言的發展



a: 說出單字

b: 說出兩個字的詞彙

c: 說出五個字以上的句子

透過與成人的接觸，兒童可以迅速且自然地習得語言，絕大部份的兒童在三歲之前，就已經能夠造出簡單的句子。

How do we study Language?

- **Linguistics (語言學)**

- The study of language structure and change

- Computational linguistics (計算語言學)

- The study of language via computational method:

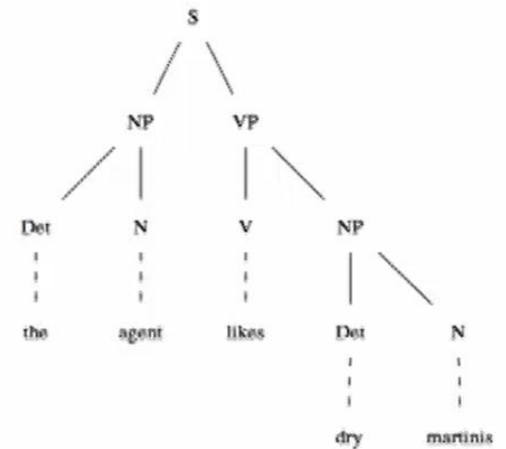
- Sociolinguistics (社會語言學)

- The study of the relationship between social behavior and language

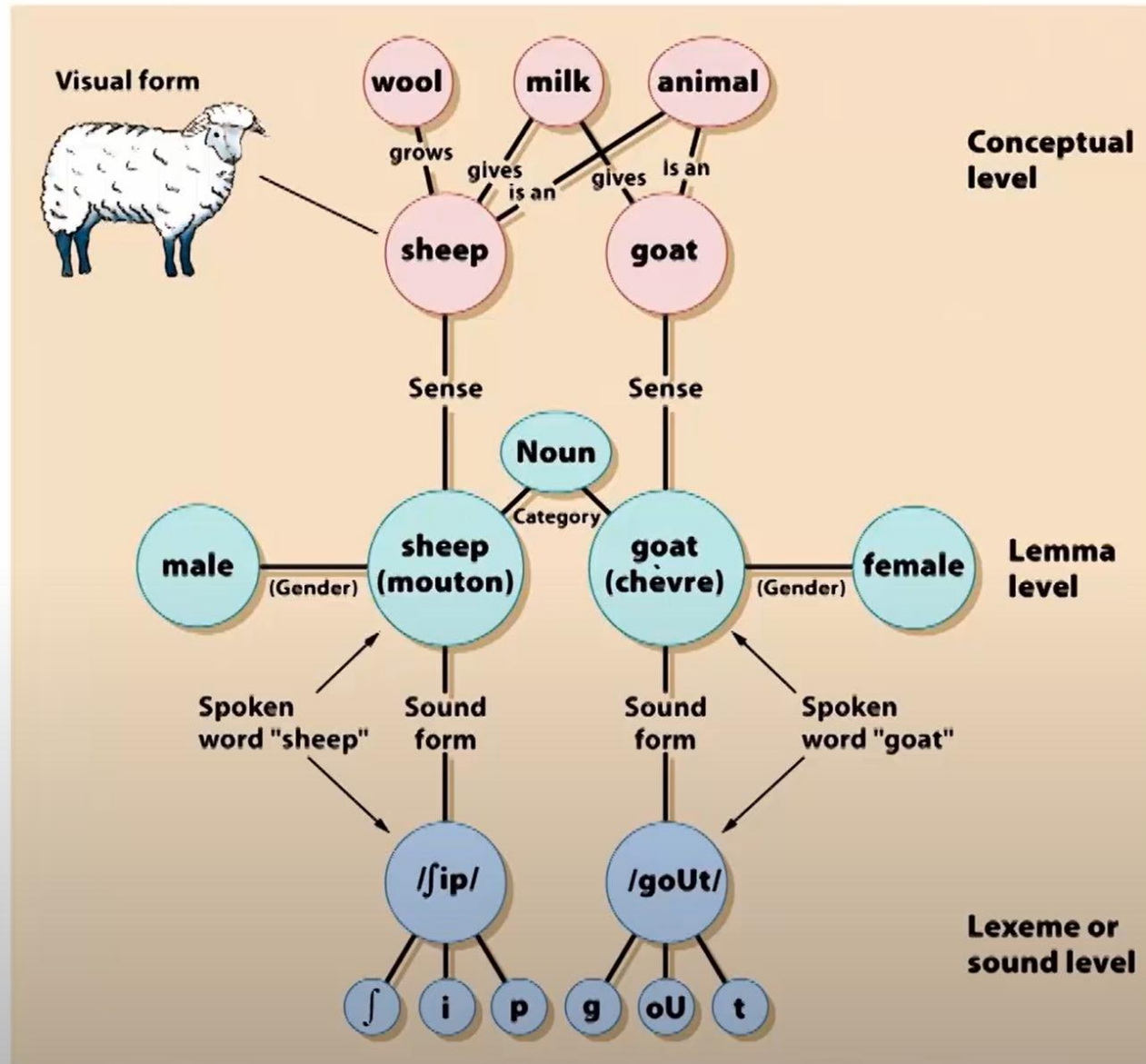
- **Psycholinguistics/ Neurolinguistics**

(心理語言學/神經語言學)

- The study of the relationships among the brain, cognition, and language



Levelt Model



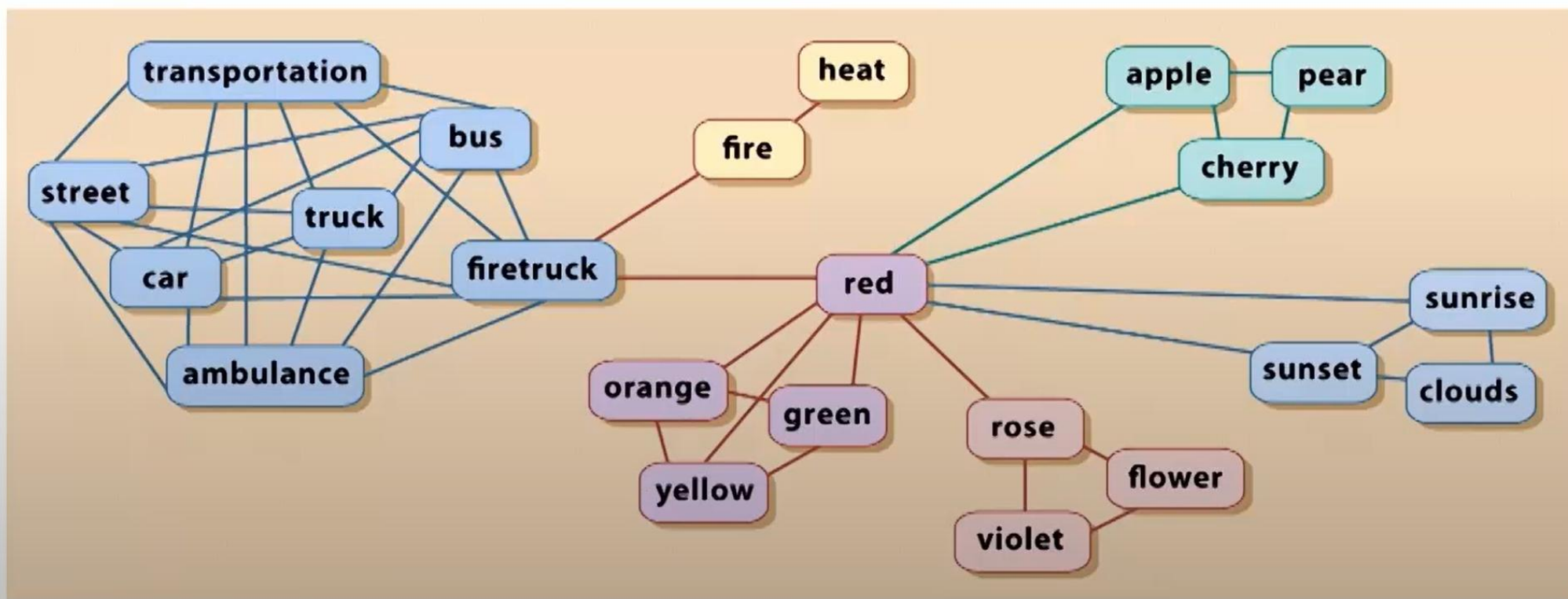
Semantic knowledge

Grammatical properties of words

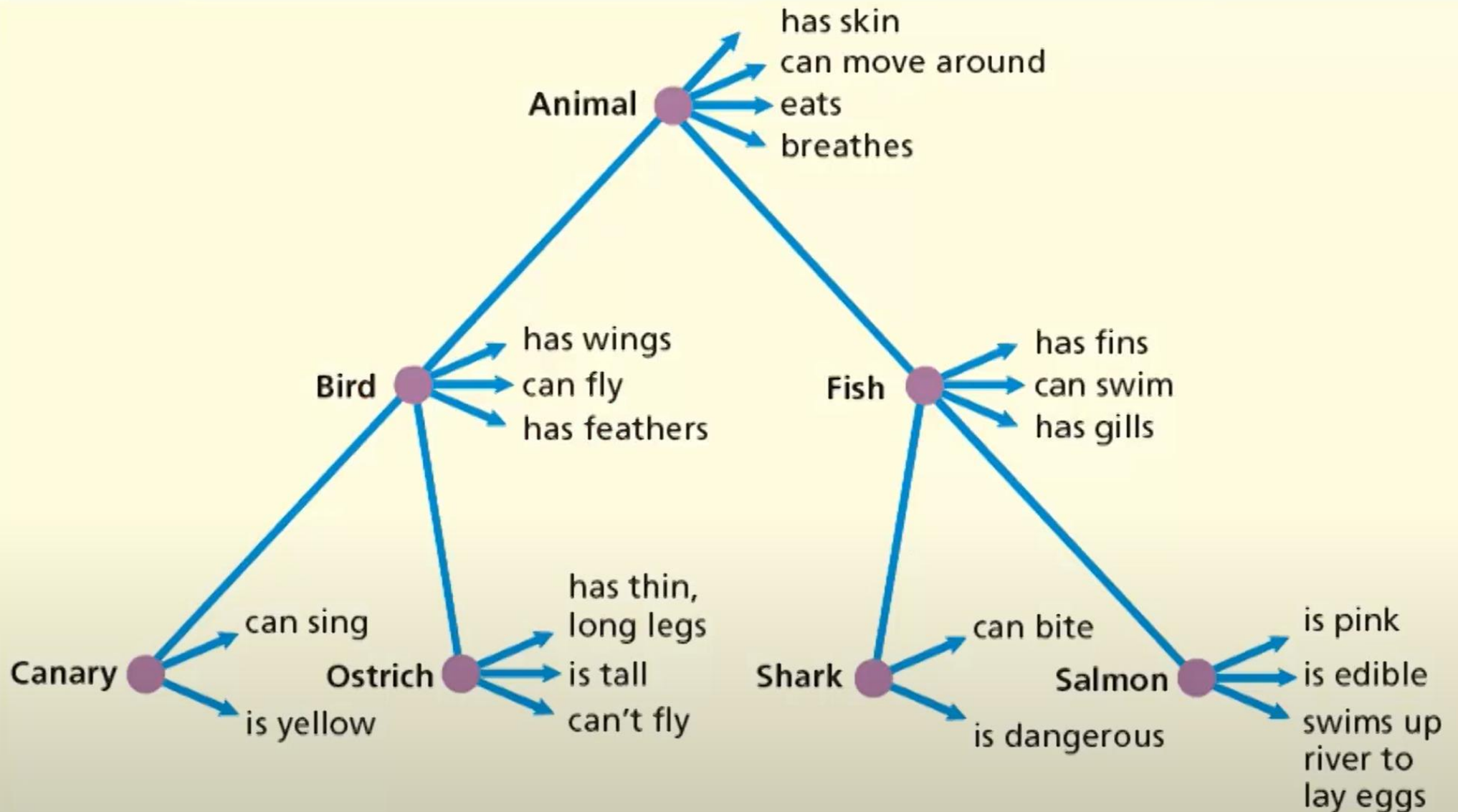
Sound patterns or Word forms

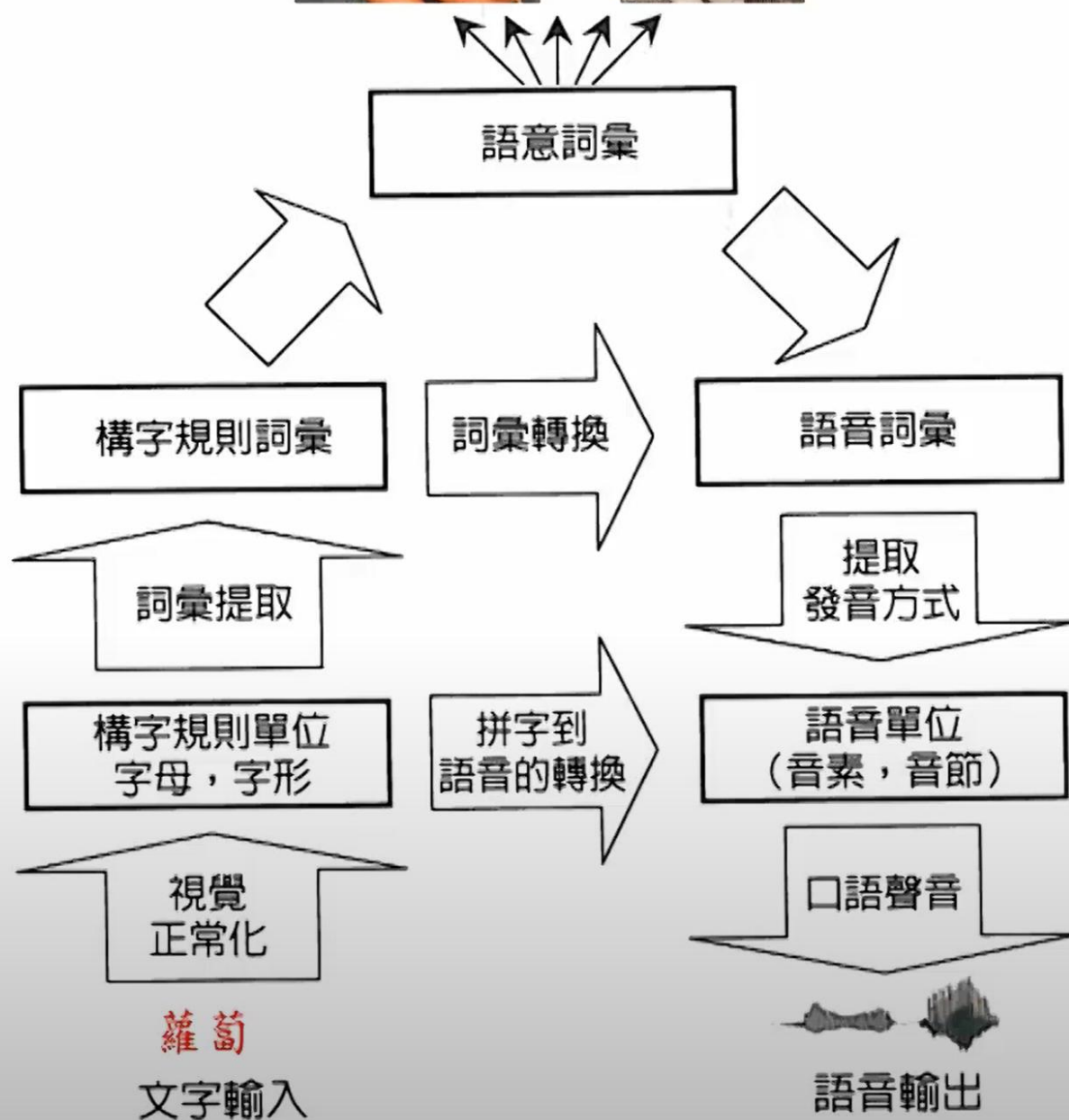
Semantic Network

- Semantic priming (語意觸接):
 - e.g., car-truck car-sunset; doctor-nurse, doctor-apple
- Distance between nodes are determined by the semantic relations between words

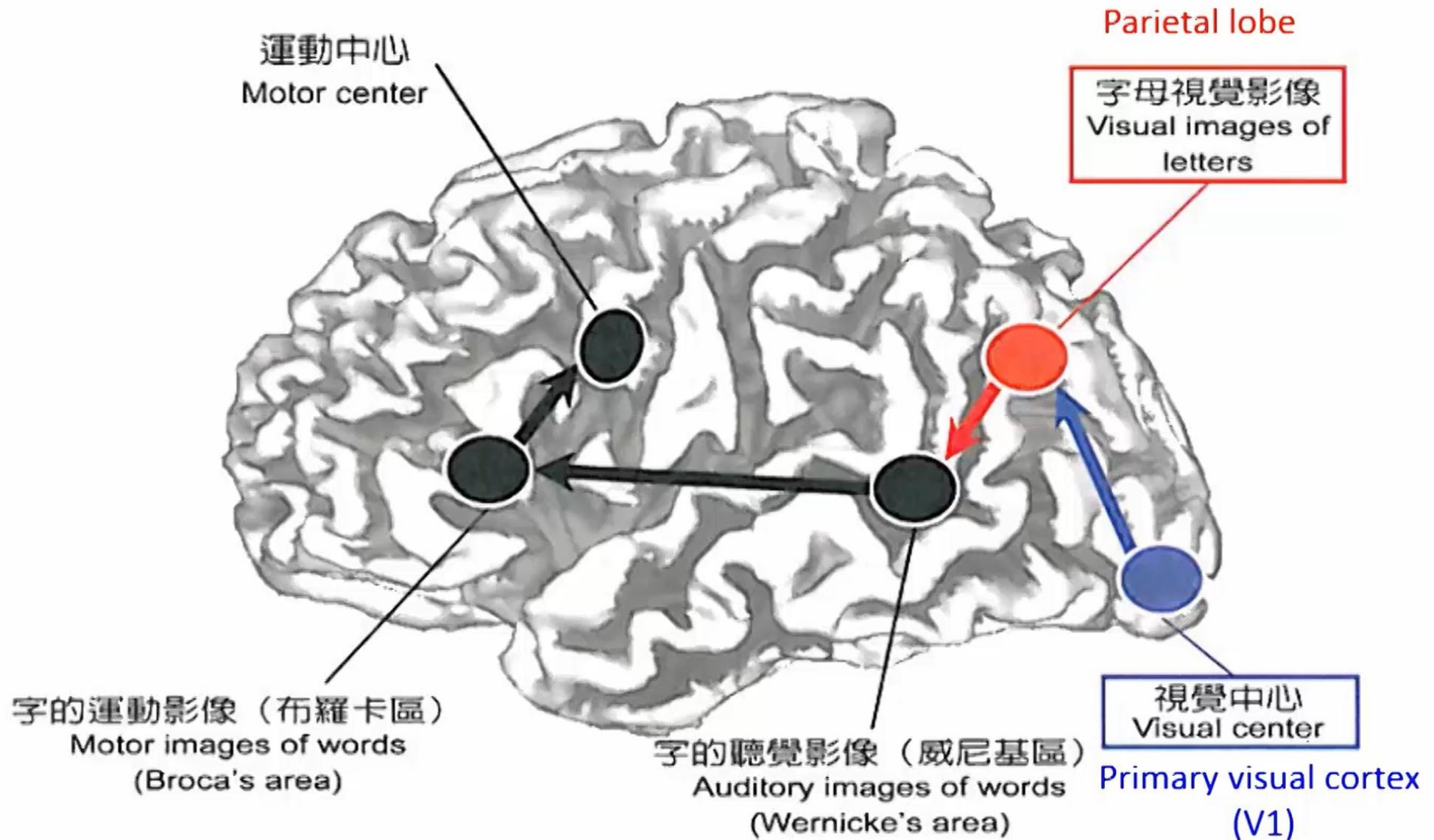


Hierarchical Model



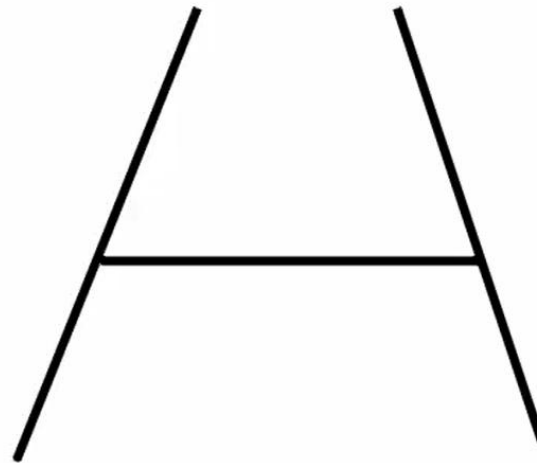


The old neurological model of reading (Geschwind, 1965)



Bottom-up and top-down: example

- What letter is this?



- Now, what letter is it?

TAE CAT

Example

According to a research at an English university, it doesn't matter in what order the letters in a word are, the only important thing is that the first and last letter be at the right place. The rest can be a total mess and you can still read it without problem. This is because we do not read every letter by itself but the word as a whole.

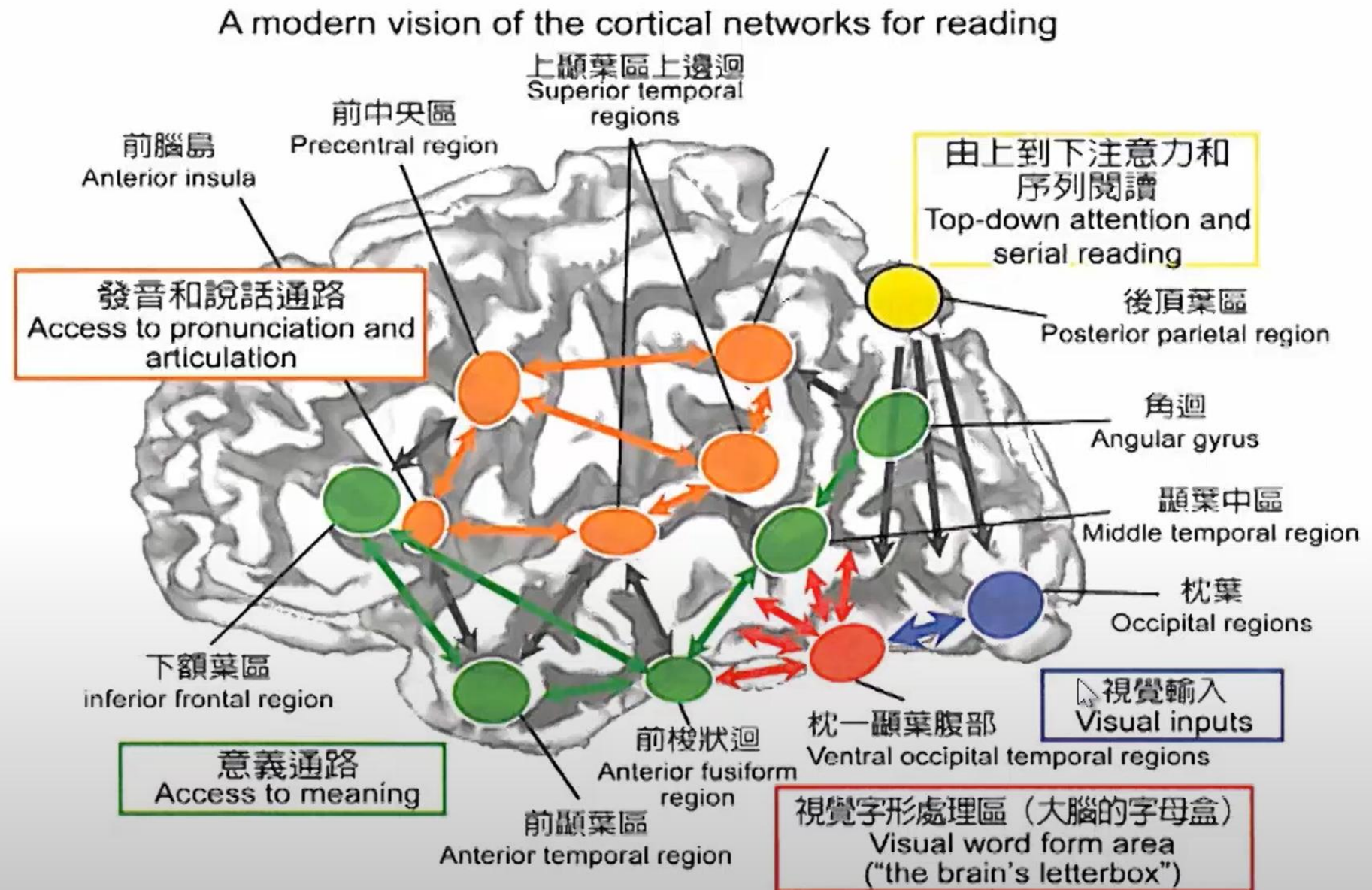
Example

研表究明，漢字的序順
並不一定一會影閱響讀。
比如當看你完這句話後
，才發這現裡的字全是
都亂的。

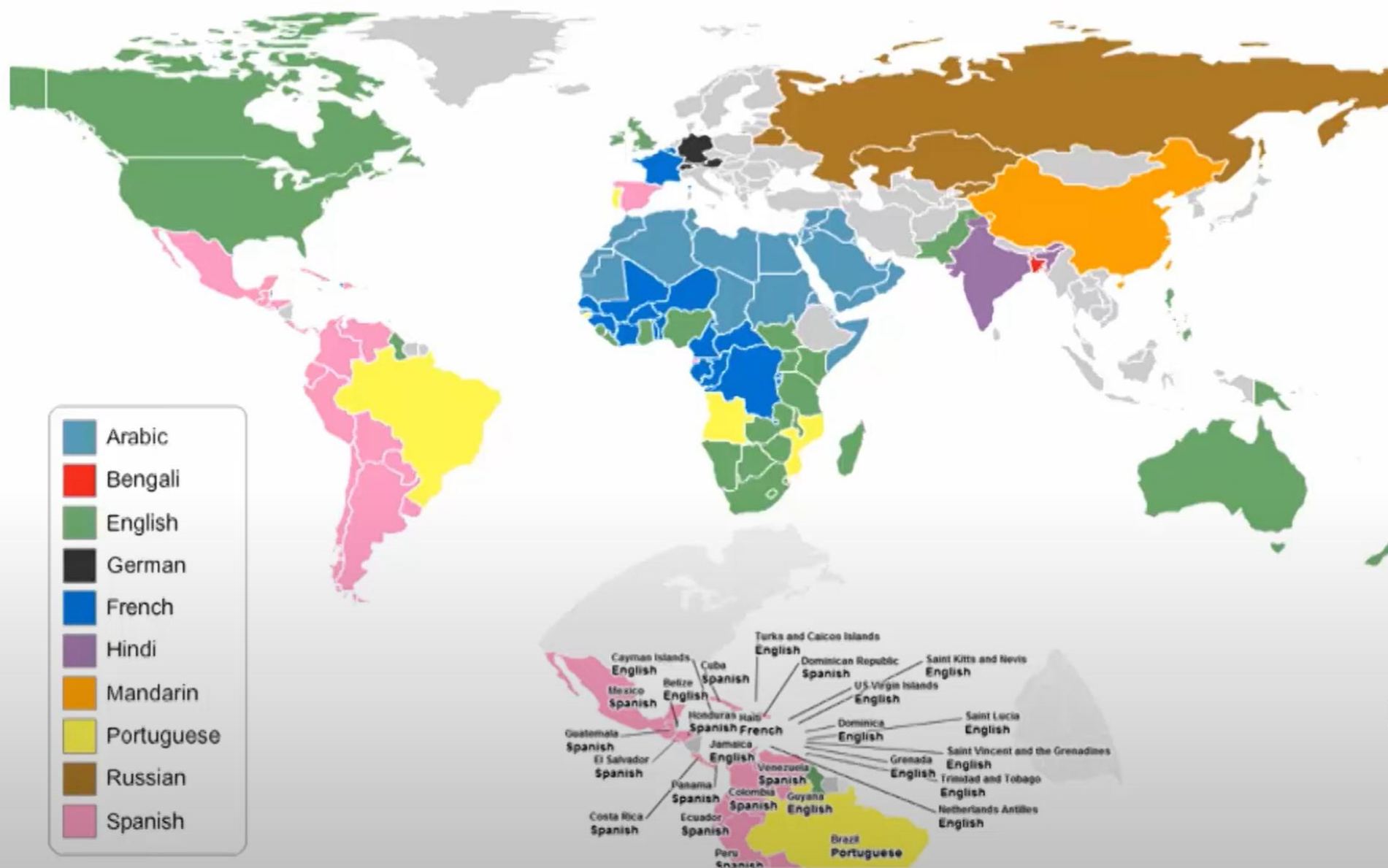
Keep in mind

- **Visual analysis** is the first step in reading.
- Subsequently, the **roots of words, meaning, sound patterns, motor articulation** must be brought into contact.
- Each of these operations typically demands the **simultaneous activation** of several separate brain regions whose connections are not organized in linear chains.
- All the connections are **bidirectional**: when a region A connects to a region B, the converse projection from B to A also exists.

Modern view of the neural network for reading



Diversity of written language



Diversity of written language

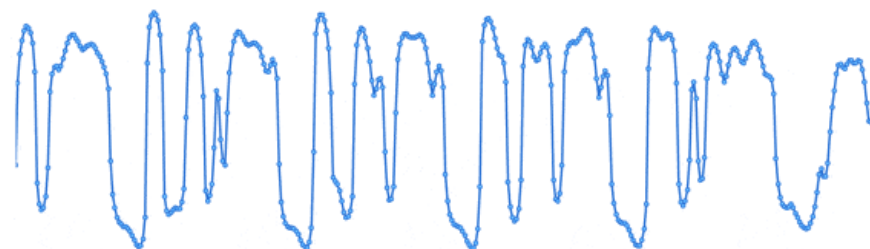
施氏食獅史

石室詩士施氏，嗜獅，誓食十獅。
施氏時時適市視獅，十時，適十
獅適市。是時，適施氏適市。施氏
視是十獅，恃失勢，使是十獅逝
世。施氏拾是十獅屍，適石室。石
室溼，施氏使侍拭石室，石室拭，
施氏始試食是十獅屍。食時，始
識食是十獅屍是十石獅屍。試釋
是事。

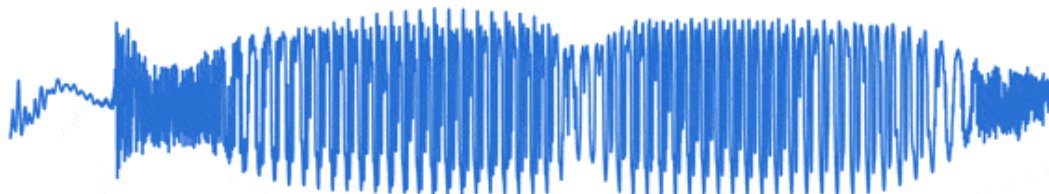
Shi Shi Shi Shi Shi Shi Shi Shi Shi
Shi Shi Shi Shi Shi Shi Shi Shi Shi
Shi Shi Shi Shi Shi Shi Shi Shi Shi
Shi Shi Shi Shi Shi Shi Shi Shi Shi
Shi Shi Shi Shi Shi Shi Shi Shi Shi
Shi Shi Shi Shi Shi Shi Shi Shi Shi
Shi Shi Shi Shi Shi Shi Shi Shi Shi
Shi Shi Shi Shi Shi Shi Shi Shi Shi
Shi Shi Shi Shi Shi Shi Shi Shi Shi
Shi Shi Shi Shi Shi Shi Shi Shi Shi



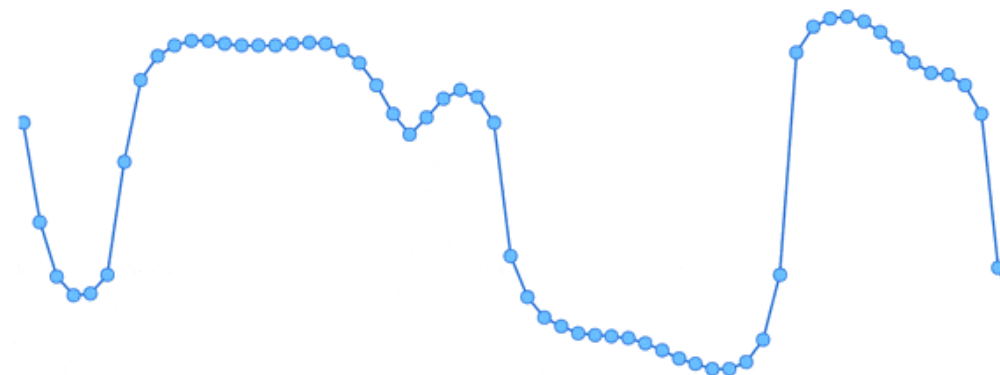
1 Second



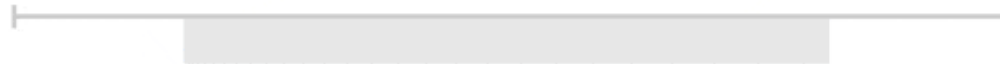
10 milliseconds



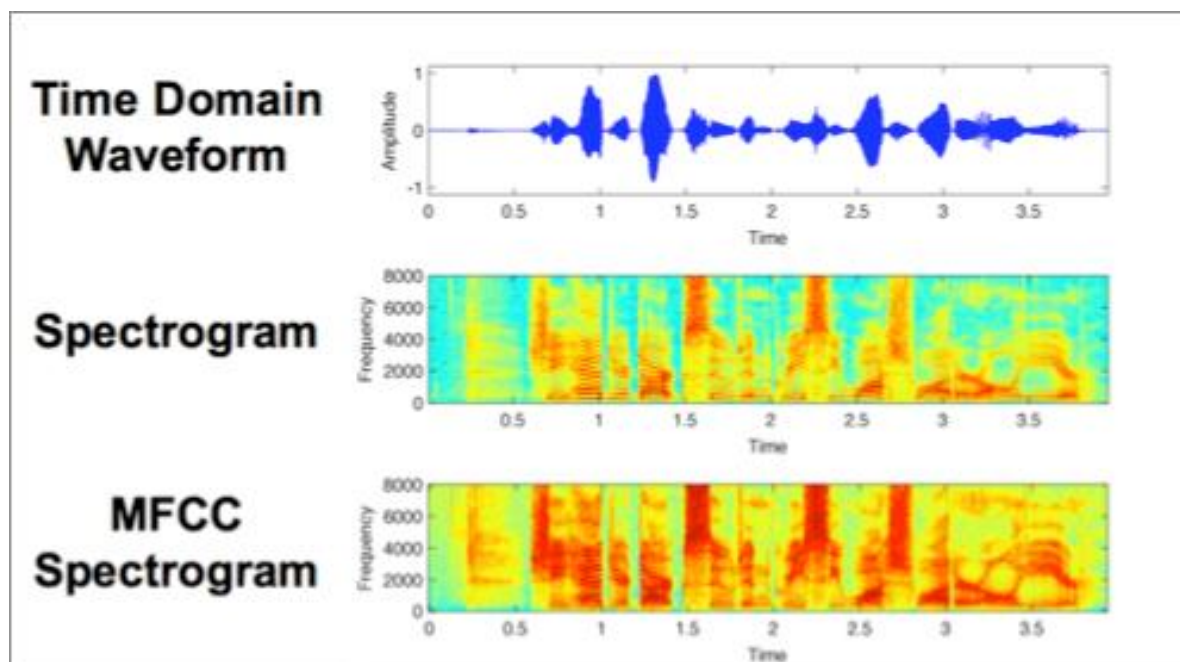
100 milliseconds



1 millisecond



Different Level Features

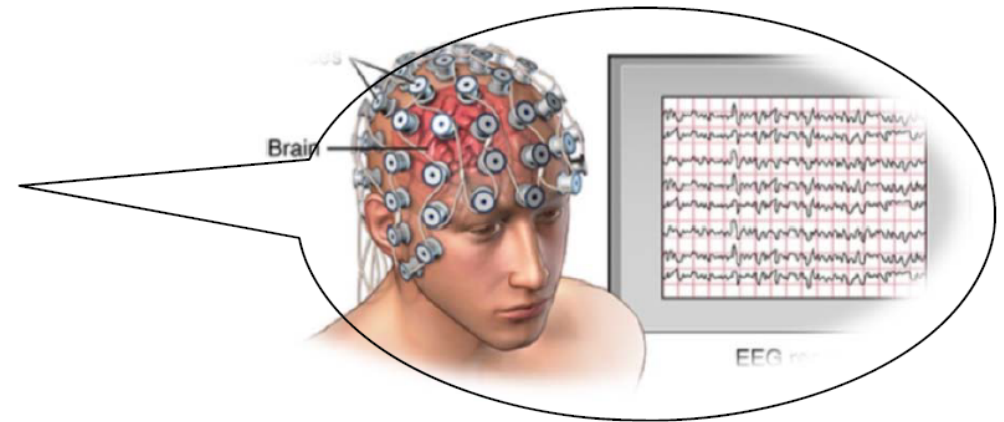


Better for deep models

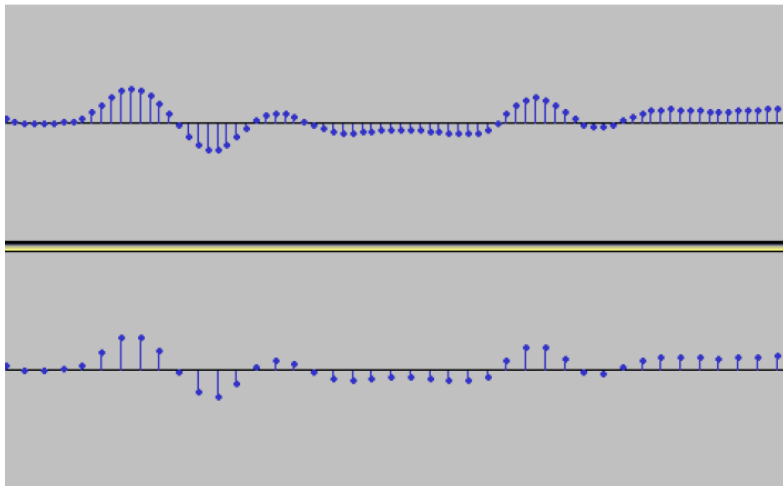
Better for shallow models

Sampling Rate

- Definition: number of samples per second
- Why: analog to digital
- Examples
 - EEG signal: **128 Hz**
 - Telephone audio: **8k Hz**
 - Music audio: **44k HZ**



<https://www.brightbraincentre.co.uk/electroencephalogram-eeeg-brainwaves/>



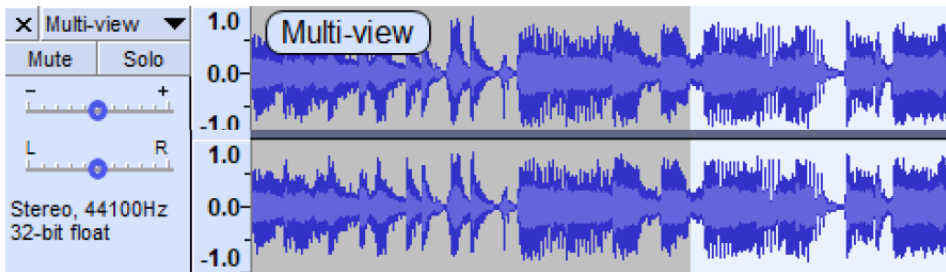
Brainwaves, Frequencies and Functions

Unconscious		Conscious		
Delta	Theta	Alpha	Beta	Gamma
0,5 – 4 Hz	4 – 8 Hz	8 – 13 Hz	13 – 30 Hz	30-42 Hz
Instinct	Emotion	Consciousness	Thought	Will
Survival Deep sleep Coma	Drives Feelings Trance Dreams	Awareness of the body Integration of feelings	Perception Concentration Mental activity	Extreme focus Energy Ecstasy

<http://altered-states.net/barry/update236/>

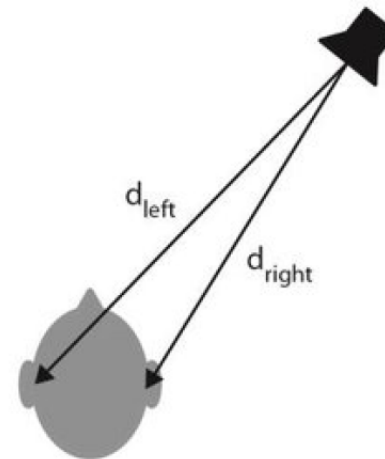
Mono vs Stereo

- Monaural
- Binaural

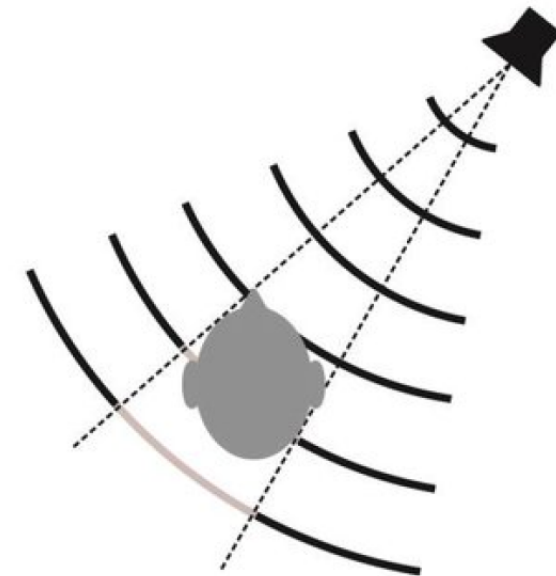


- Stereo-to-mono conversion
 - By taking the average
 - Mono-to-stereo conversion
 - Need research
- https://www.youtube.com/watch?v=aWxmQKm_s8Q

a) Binaural localization cue:
interaural time difference (ITD)



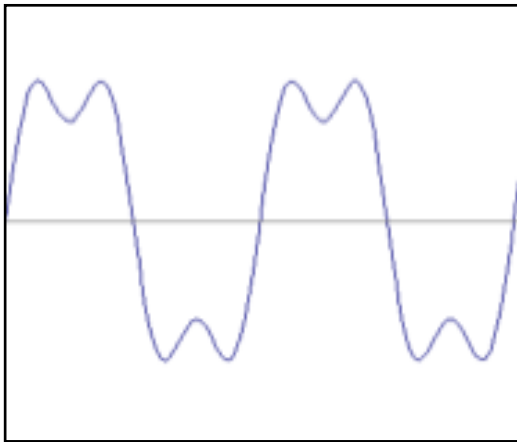
b) Binaural localization cue:
interaural intensity difference (IID)



Source: https://www.researchgate.net/figure/Binaural-and-monaural-cues-used-for-sound-localization-a-For-sound-sources-off-the_fig1_299281975

Examples

How would you generate this function?



=

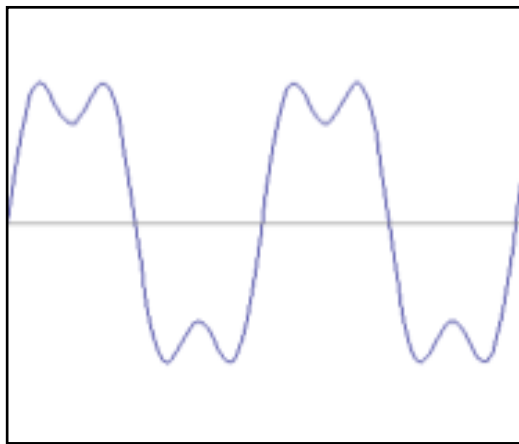
?

+

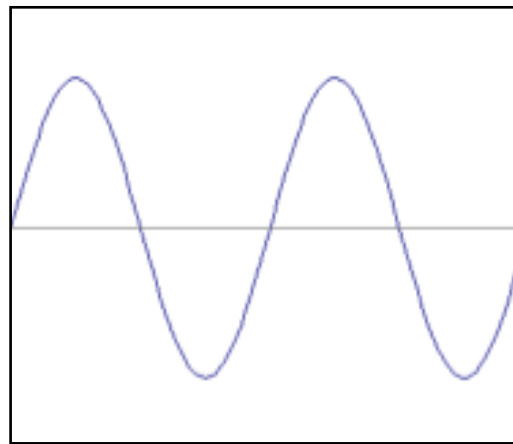
?

Examples

How would you generate this function?



=



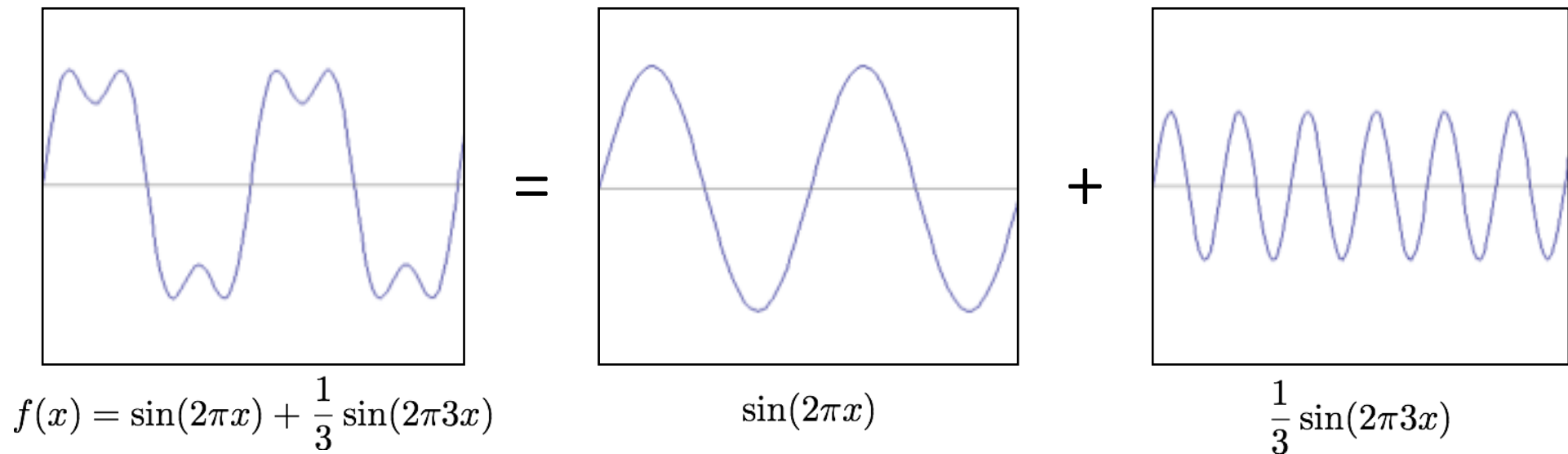
$\sin(2\pi x)$

+

?

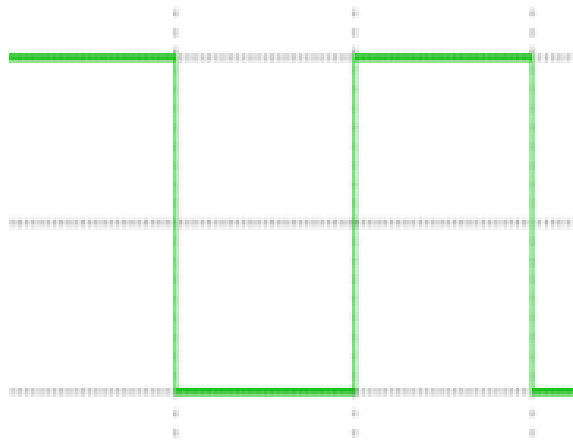
Examples

How would you generate this function?



Examples

How would you generate this function?



square wave

=

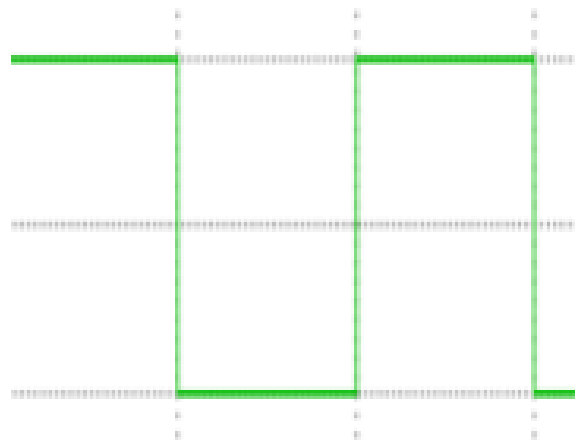
?

+

?

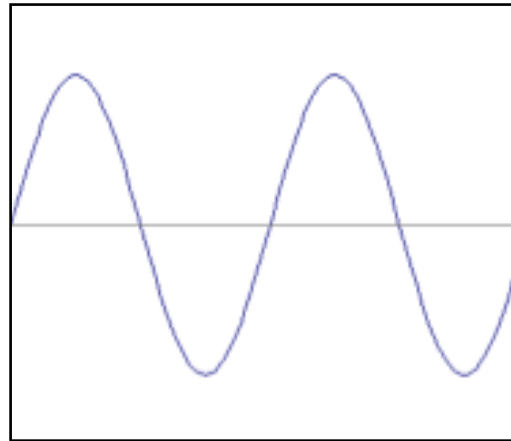
Examples

How would you generate this function?

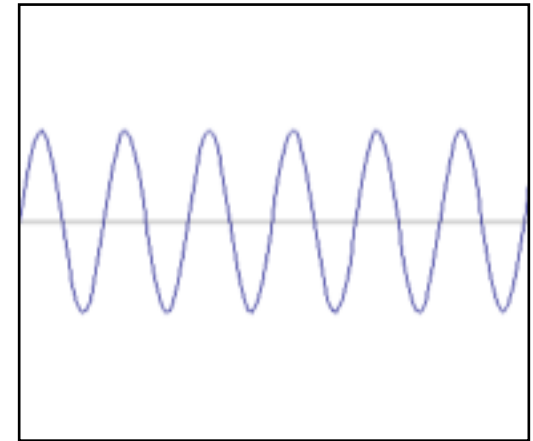


square wave

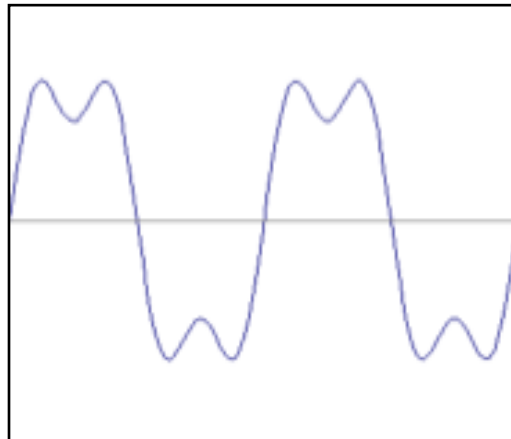
$\approx$



+

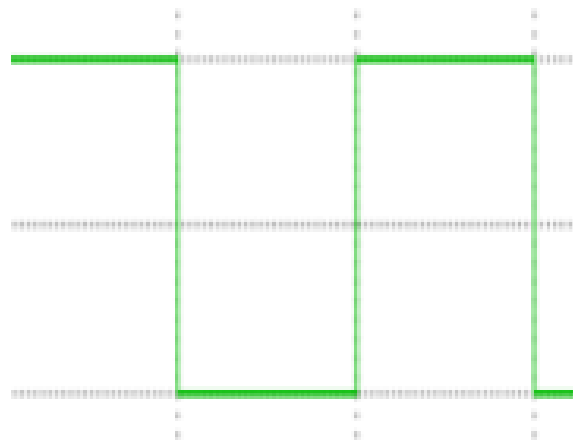


$=$



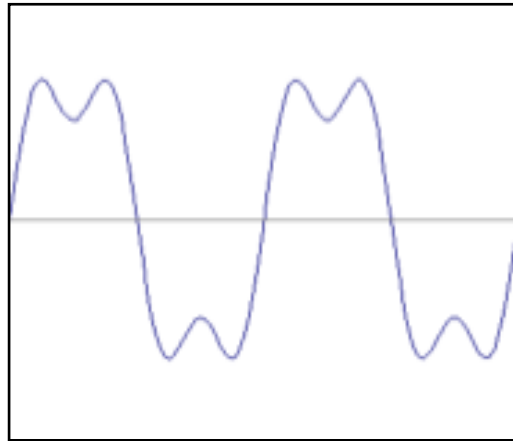
Examples

How would you generate this function?

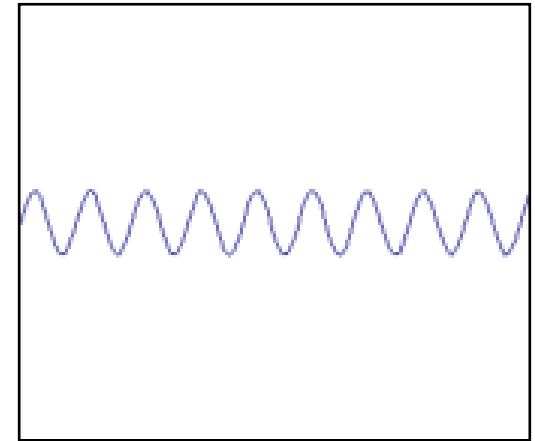


square wave

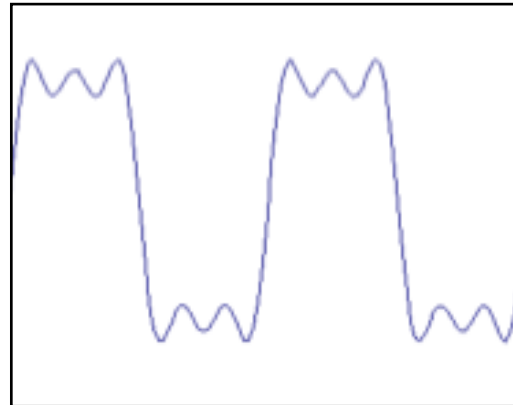
$\approx$



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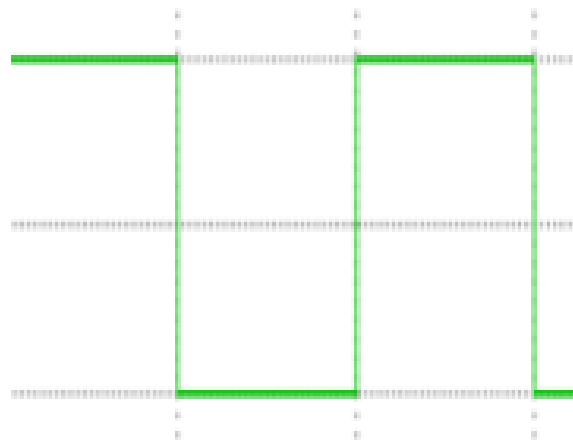


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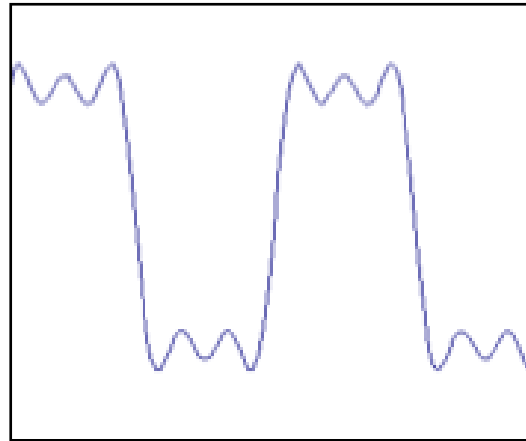
Examples

How would you generate this function?

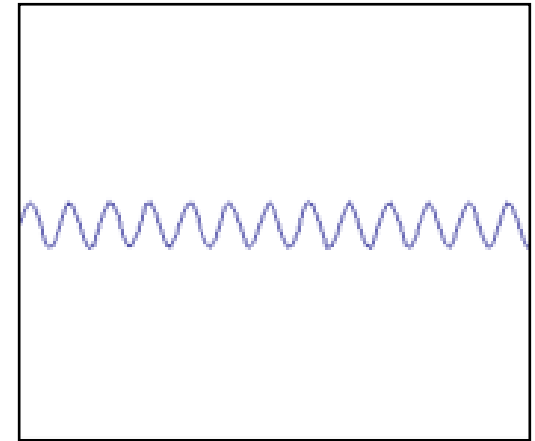


square wave

$\approx$



+

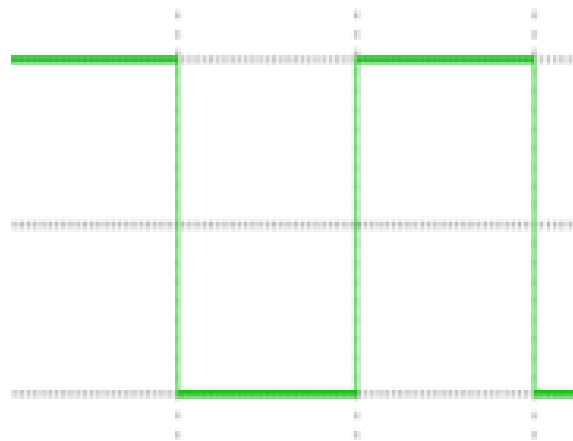


$=$



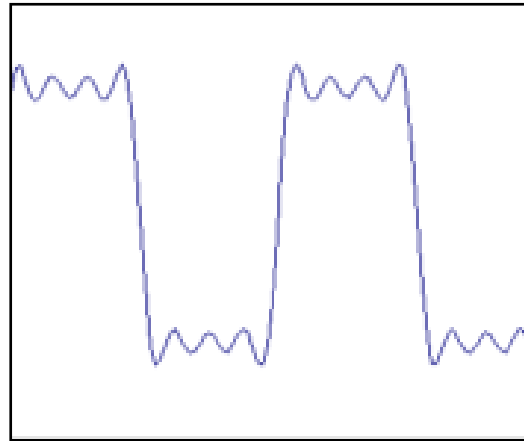
Examples

How would you generate this function?

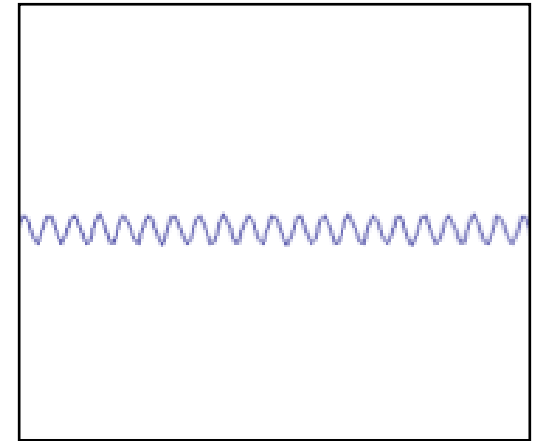


square wave

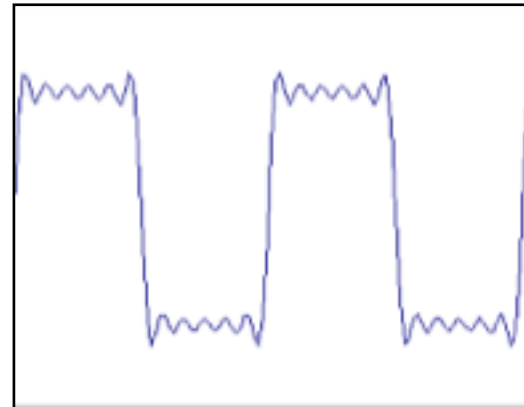
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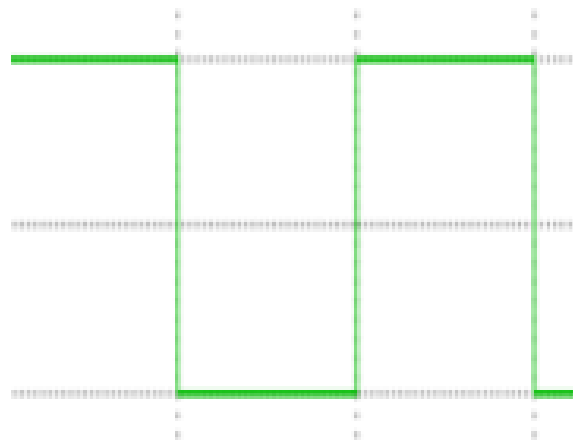


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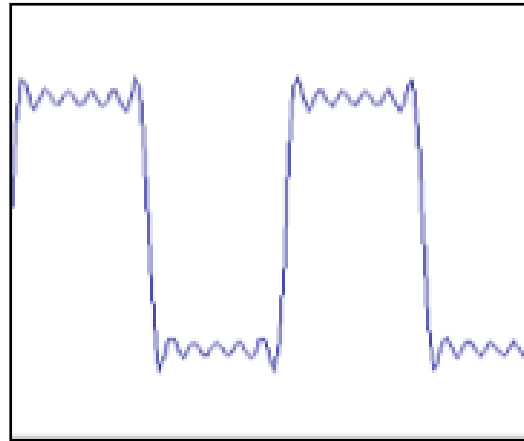
Examples

How would you generate this function?

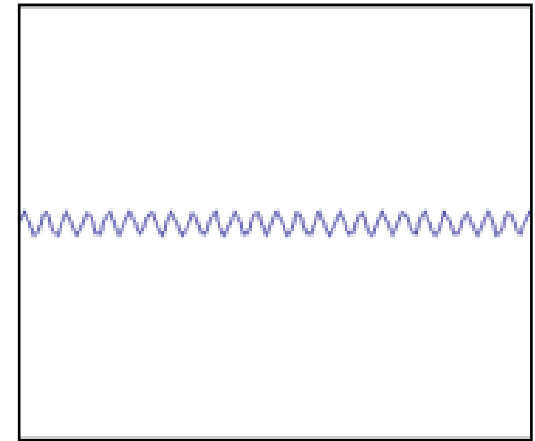


square wave

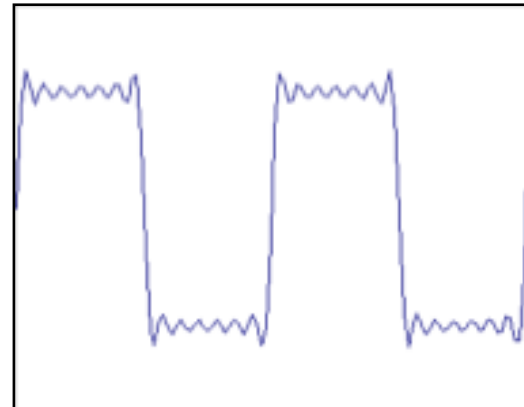
$\approx$



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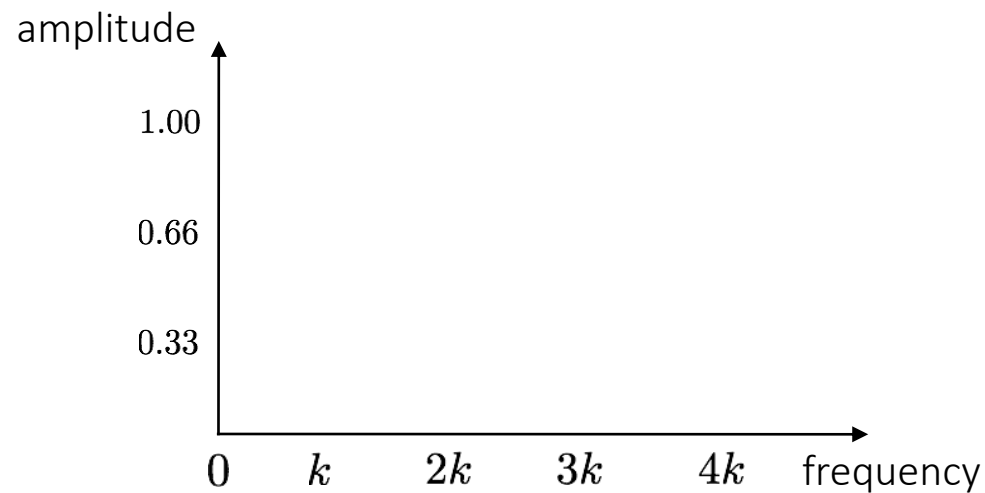


$=$



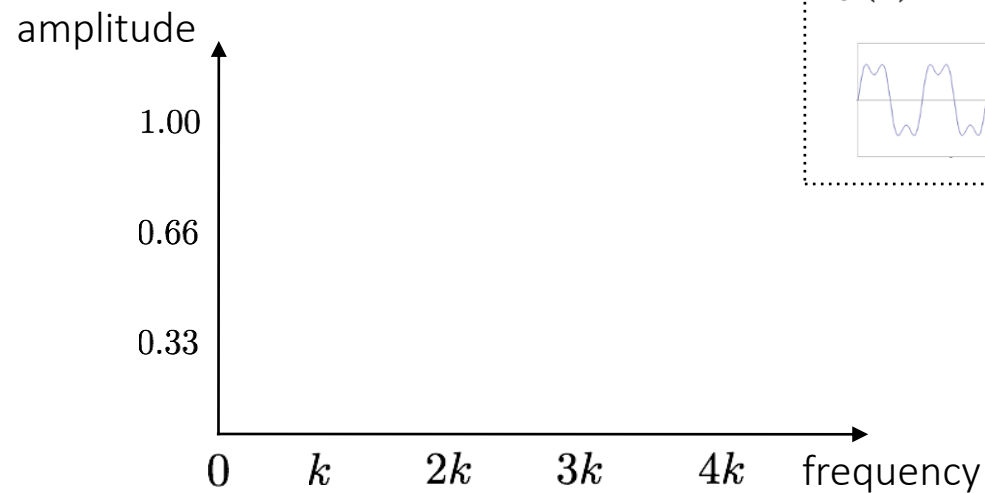
How would you express this mathematically?

Visualizing the frequency spectrum



Visualizing the frequency spectrum

Recall the temporal domain visualization



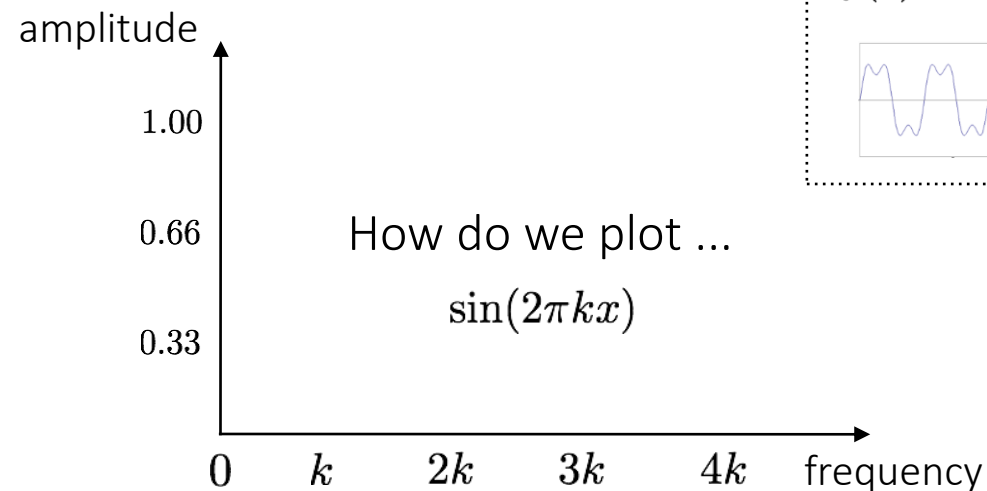
$$f(x) = \sin(2\pi kx) + \frac{1}{3} \sin(2\pi 3kx)$$

$\sin(2\pi kx) + \frac{1}{3} \sin(2\pi 3kx)$

Visualizing the frequency spectrum

Recall the temporal domain visualization

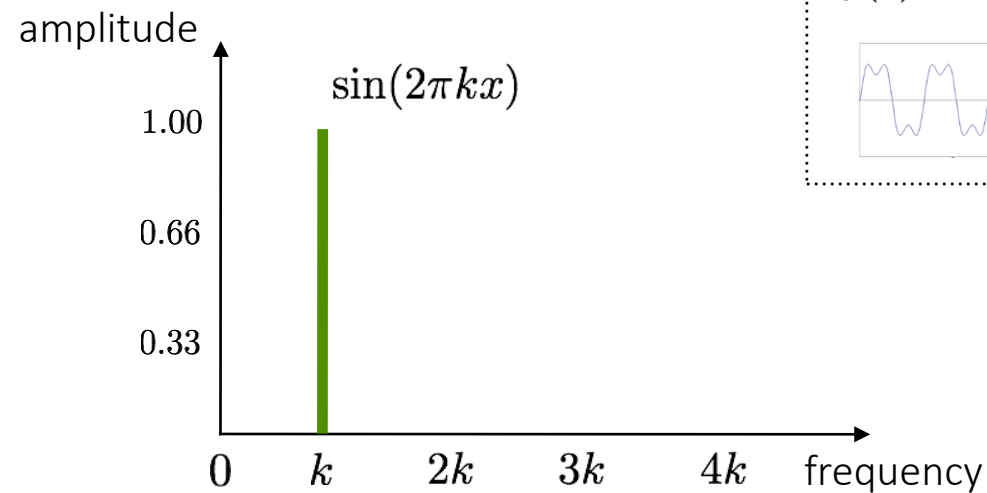
$$f(x) = \sin(2\pi kx) + \frac{1}{3} \sin(2\pi 3kx)$$

Visualizing the frequency spectrum

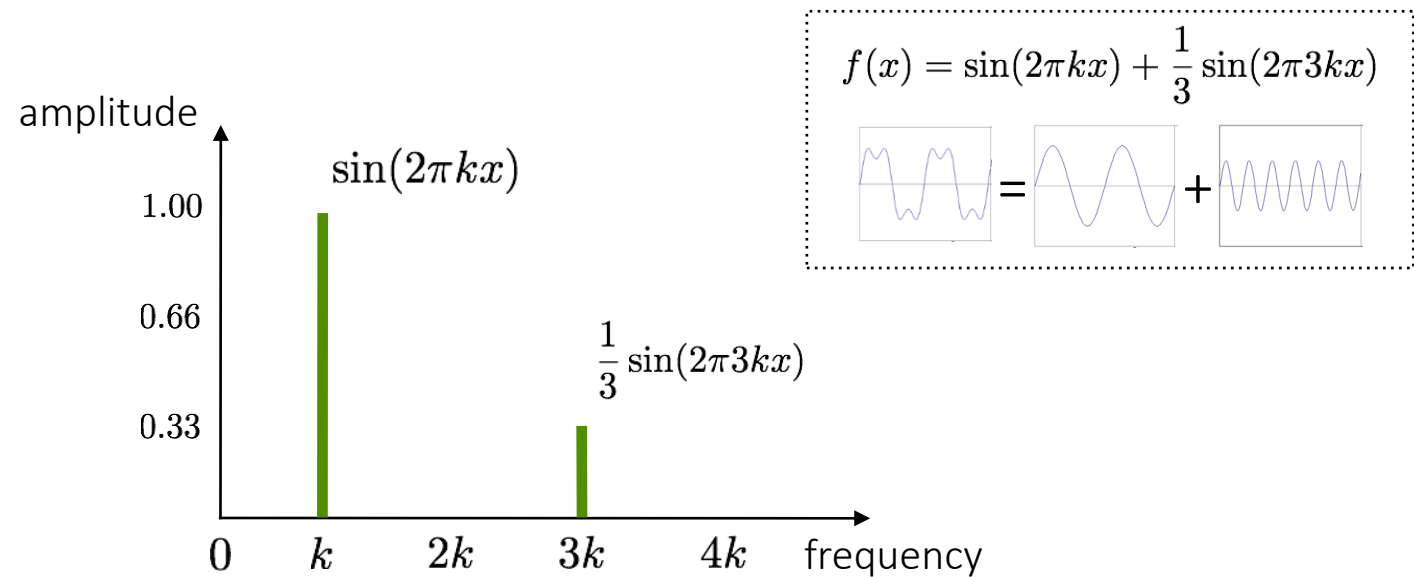
Recall the temporal domain visualization

$$f(x) = \sin(2\pi kx) + \frac{1}{3} \sin(2\pi 3kx)$$

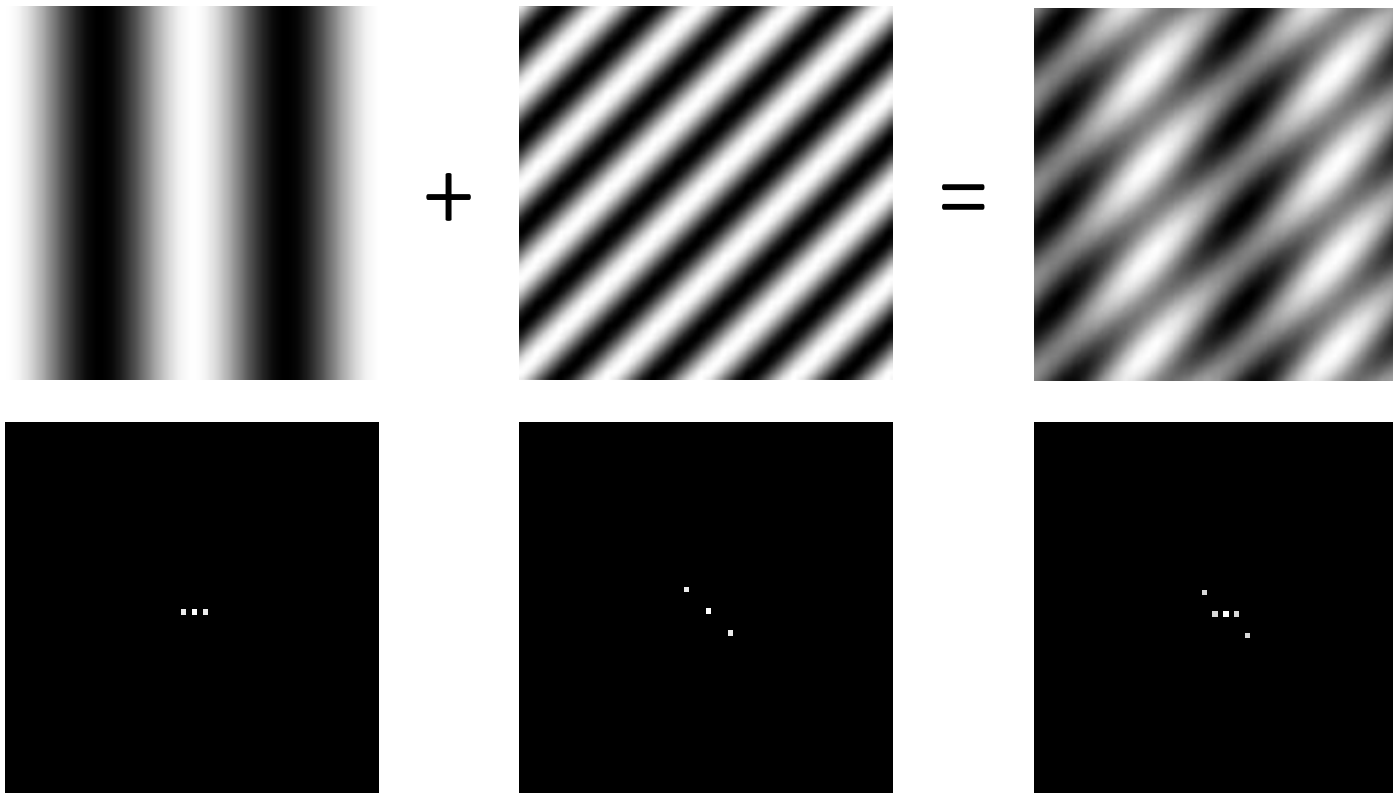


Visualizing the frequency spectrum

Recall the temporal domain visualization



Examples



$$f = 2^{(d-69)/12} \cdot 440 \text{ Hz} \qquad d = 69 + 12 \log_2 \left(\frac{f}{440 \text{ Hz}} \right)$$

Notes and Pitches

(Assuming A4=440 Hz; but there are other *tunings*)

<https://newt.phys.unsw.edu.au/jw/notes.html>

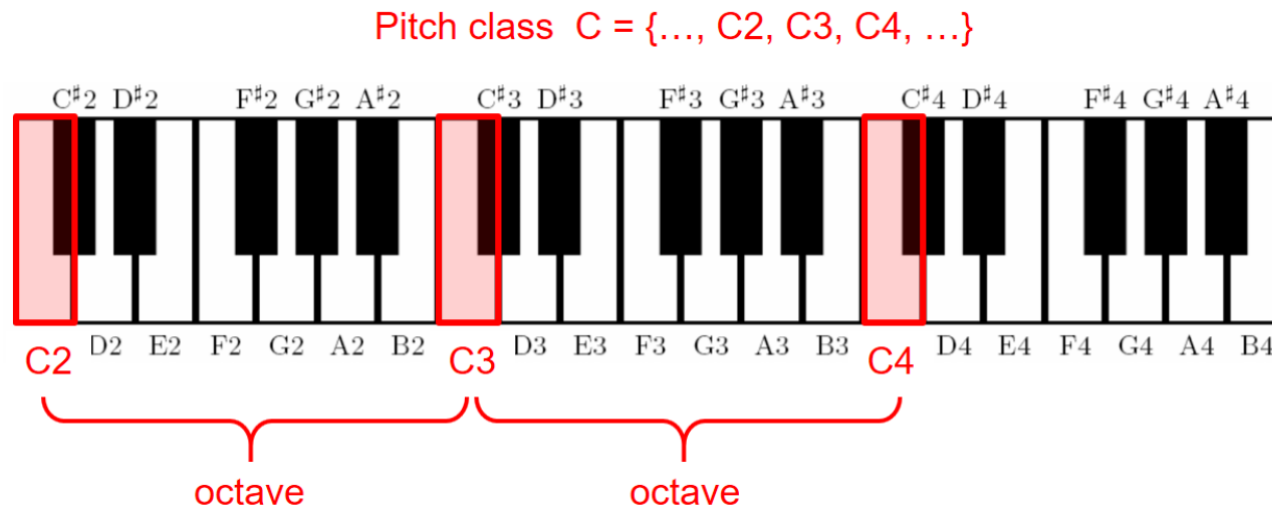
Note name	Keyboard	Frequency Hz	
A0		27.500	
B0		30.868	29.135
C1		32.703	
D1		36.708	34.648
E1		41.203	38.891
F1		43.654	
G1		48.999	46.249
A1		55.000	51.913
B1		61.735	58.270
C2		65.406	
D2		73.416	69.296
E2		82.407	77.782
F2		87.307	
G2		97.999	92.499
A2		110.00	103.83
B2		123.47	116.54
C3		130.81	
D3		146.83	138.59
E3		164.81	155.56
F3		174.61	
G3		196.00	185.00
A3		220.00	207.65
B3		246.94	233.08
C4		261.63	
D4		293.67	277.18
E4		329.63	311.13
F4		349.23	
G4		392.00	369.99
A4		440.00	415.30
B4		493.88	466.16
C5		523.25	
D5		587.33	554.37
E5		659.26	622.25
F5		698.46	
G5		783.99	739.99
A5		880.00	830.61
B5		987.77	932.33
C6		1046.5	
D6		1174.7	1108.7
E6		1318.5	1244.5
F6		1396.9	
G6		1568.0	1480.0
A6		1760.0	1661.2
B6		1975.5	1864.7
C7		2093.0	
D7		2349.3	2217.5
E7		2637.0	2489.0
F7		2793.0	
G7		3136.0	2960.0
A7		3520.0	3322.4
B7		3951.1	3729.3
C8		4186.0	

Notes and Pitches

https://www.audiolabs-erlangen.de/resources/MIR/FMP/C1/C1S1_MusicalNotesPitches.html

- **Pitch Class**

- Two notes with *fundamental frequencies* in a ratio equal to *any power of two* (e.g., half, twice, or four times) are perceived as very **similar**
- All notes with this kind of relation can be grouped under the same *pitch class*



Pitch

- Pitch (音高): 構成樂音的最基本要素在於音高，也就是聲音的頻率。
- 在樂理上，樂音音符可分為七個基本音，即【Do Re Me Fa Sol La Si】，以美式的符號則記為【C D E F G A B】而第八個音則稱為高八度的Do。

Octave

- 在數理上，每一個八度音程(Octave)正好對應於不同的振動模式，而兩個八度音程差的音在頻率上正好差上兩倍。例如：在第0個八度的La(記為A0)頻率為27.5 Hertz，則第1個八度的La(記為A1)頻率即為 $27.5 \times 2 = 55.0$ Hertz。在這每一個八度的音程中，又可再將其等分為12個頻率差相近的音，這分別對應於【C Db D Eb E F Gb G Ab A Bb B】，這樣的等分法就是所謂的十二平均律(Twelve-Tone Scale)。這當中每一個音符所對應的頻率，都可以藉由數學的方程式準確的算出

Timbre: Different Instruments

https://www.audiolabs-erlangen.de/resources/MIR/FMP/C1/C1S3_Timbre.html

- **Timbre** (tone color; tone quality)
 - Allows a listener to distinguish the musical tone of a violin, an oboe, or a trumpet even if the tone is played at the same **pitch** and with the same **loudness**
 - Hard to grasp & *subjective*
 - May be described with words such as *bright*, *dark*, *warm*, and *harsh*

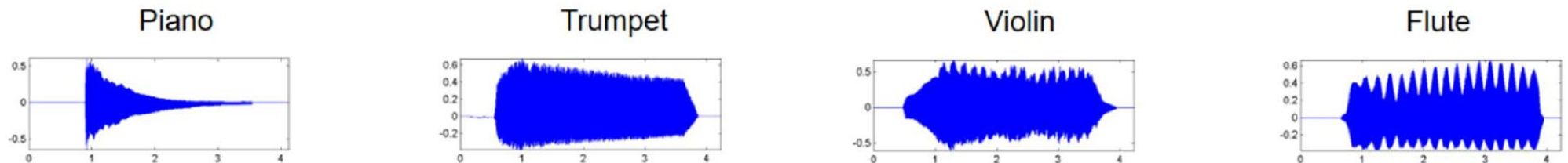


Figure 1.23 from [Müller, FMP, Springer 2015]

▶ 0:04 / 0:04 — 🔊

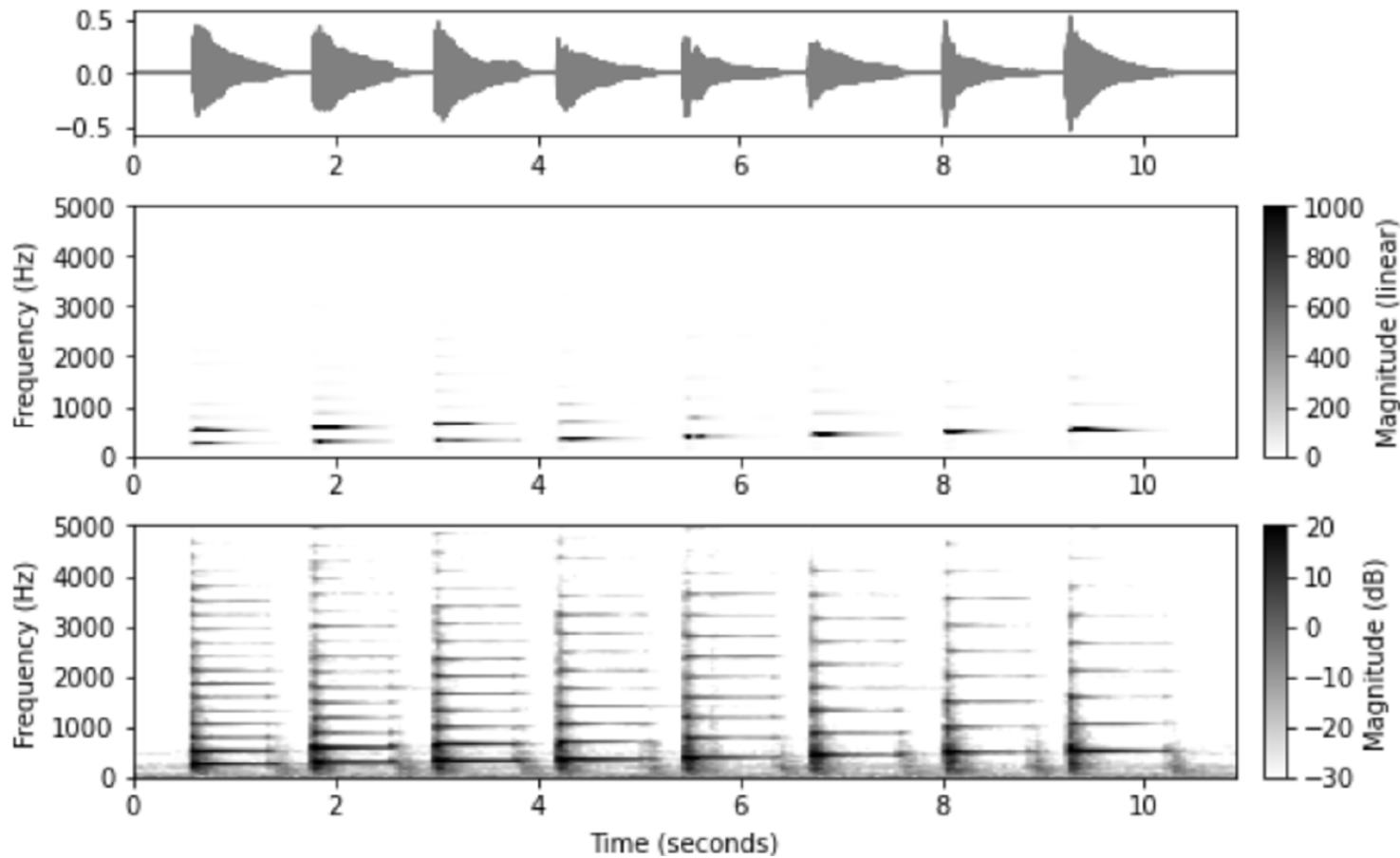
▶ 0:00 / 0:04 — 🔊

▶ 0:00 / 0:04 — 🔊

▶ 0:00 / 0:04 — 🔊

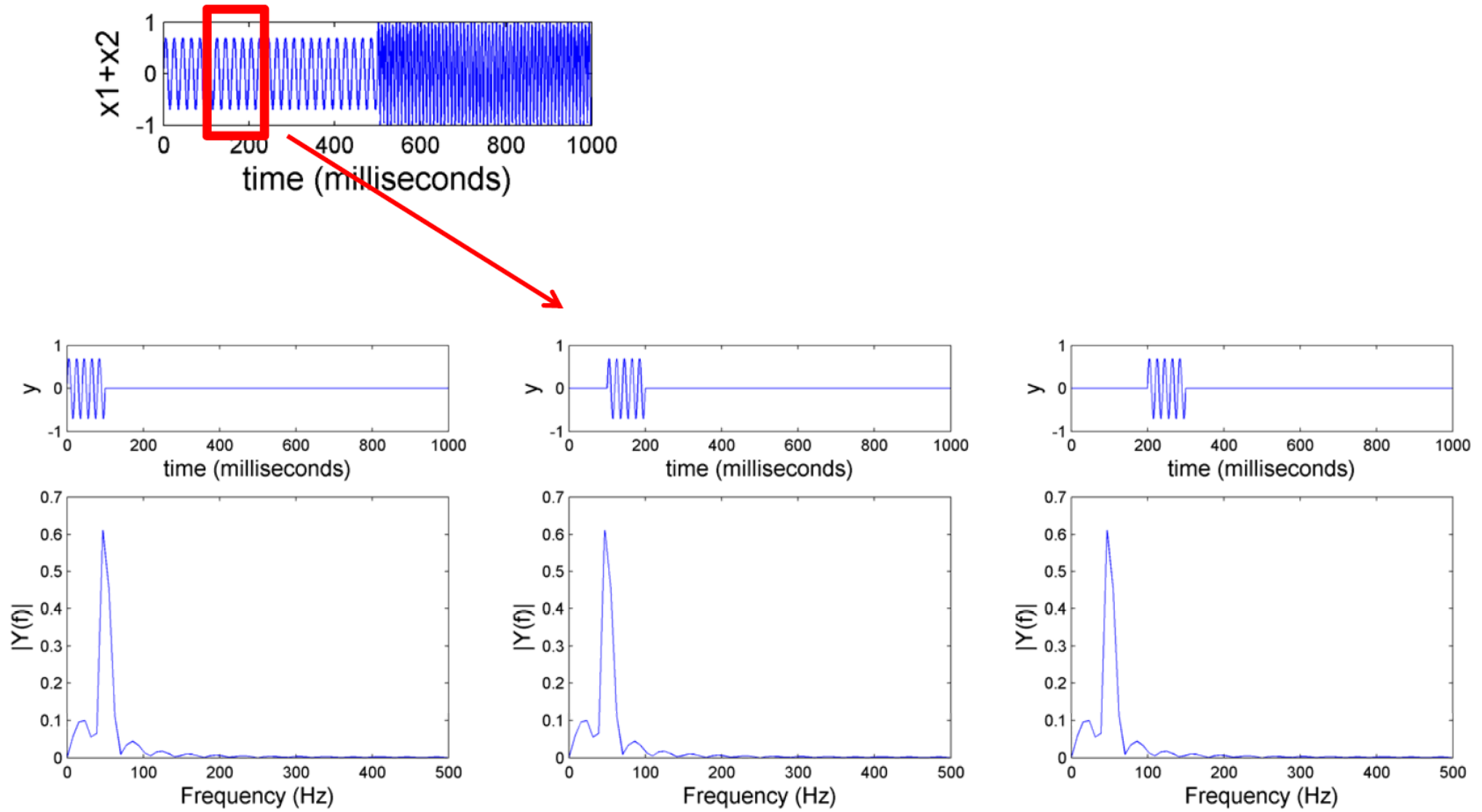
Discrete Short-Time Fourier Transform

https://www.audiolabs-erlangen.de/resources/MIR/FMP/C2/C2_STFT-Basic.html

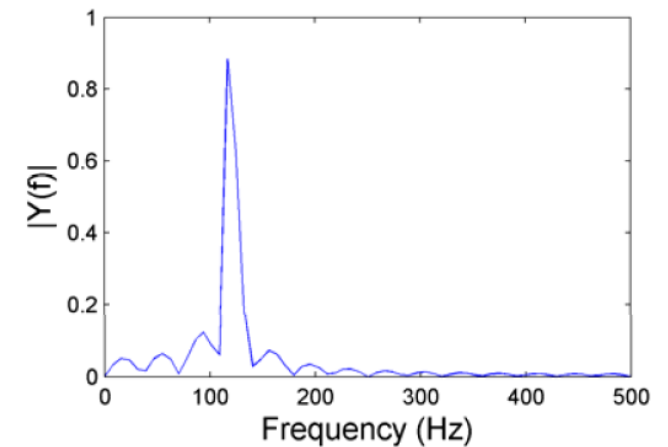
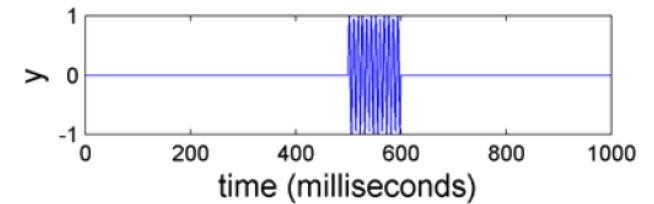
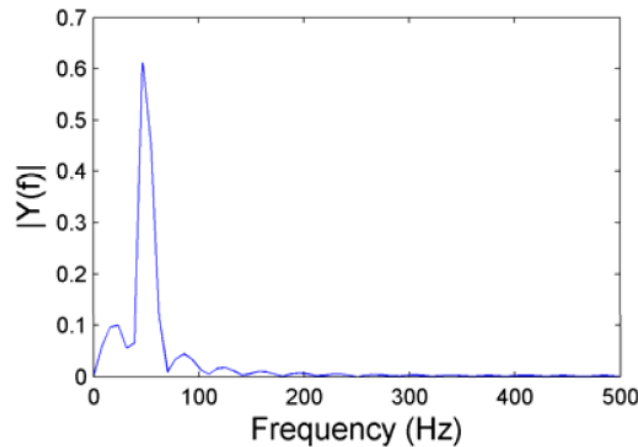
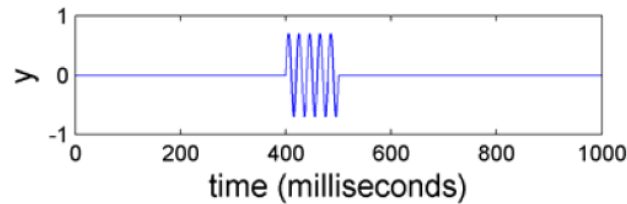
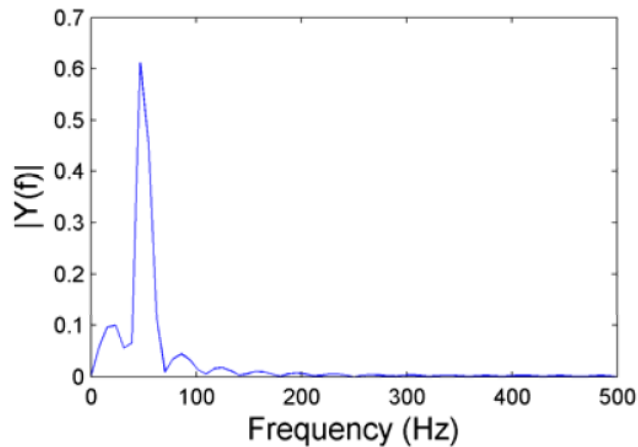
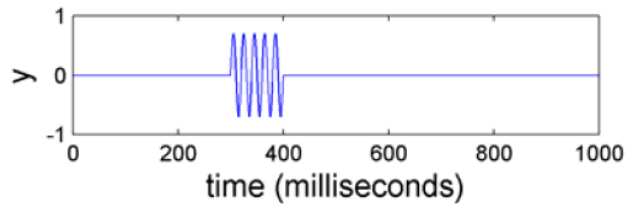
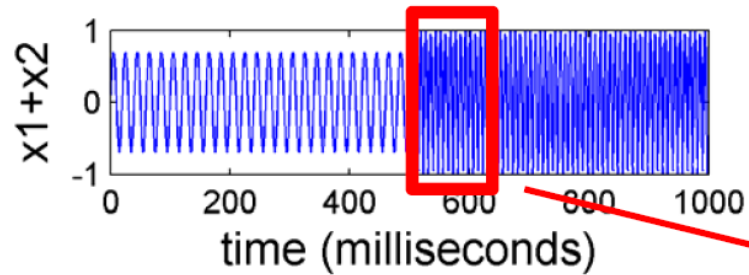


- **Time-frequency representation of audio**
 - Linear frequency, **linear** magnitude
 - Linear frequency, **logarithmic** magnitude (dB)

Short Time Fourier Transform (STFT)



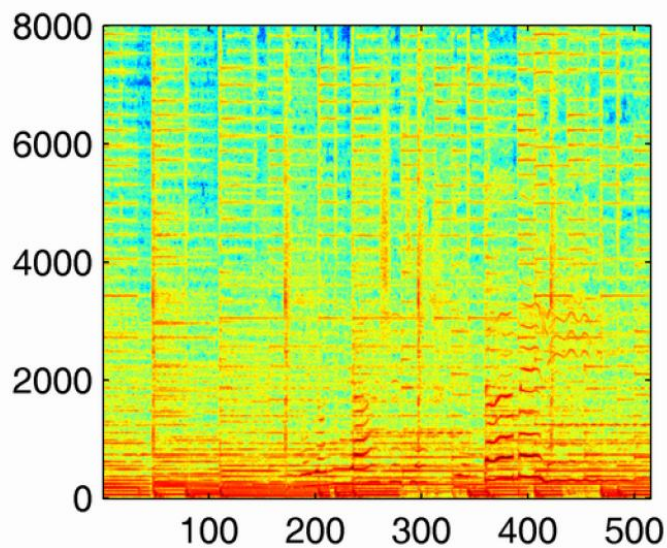
Short Time Fourier Transform (STFT)



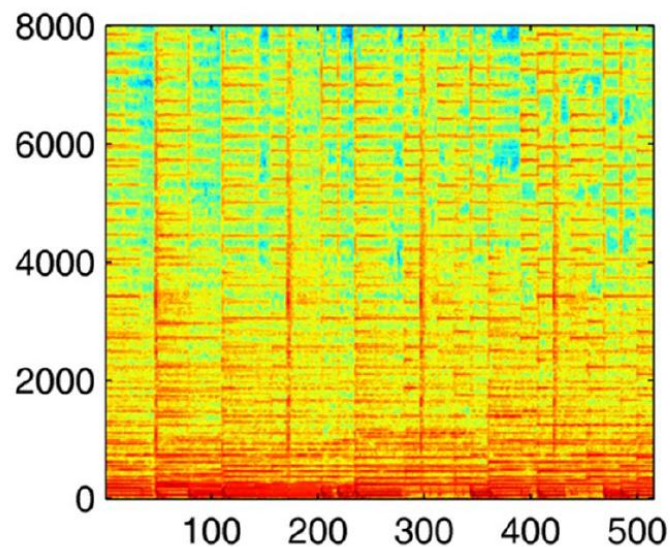
Discrete Short-Time Fourier Transform

- Time-frequency representation of audio

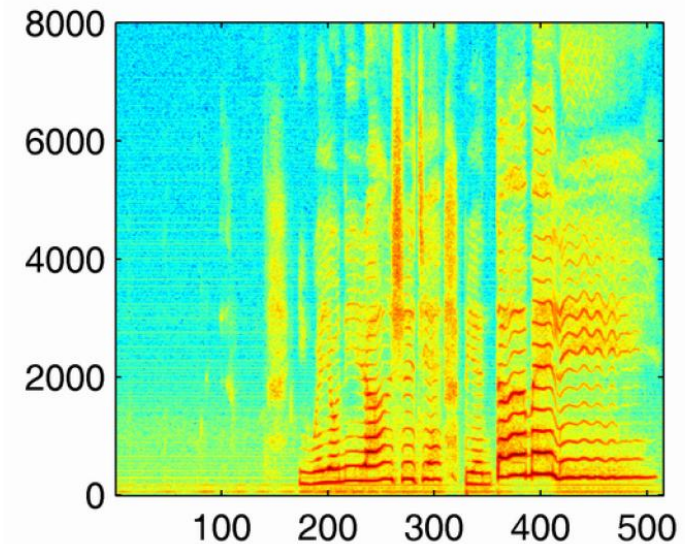
mixture



instrumental (clean)



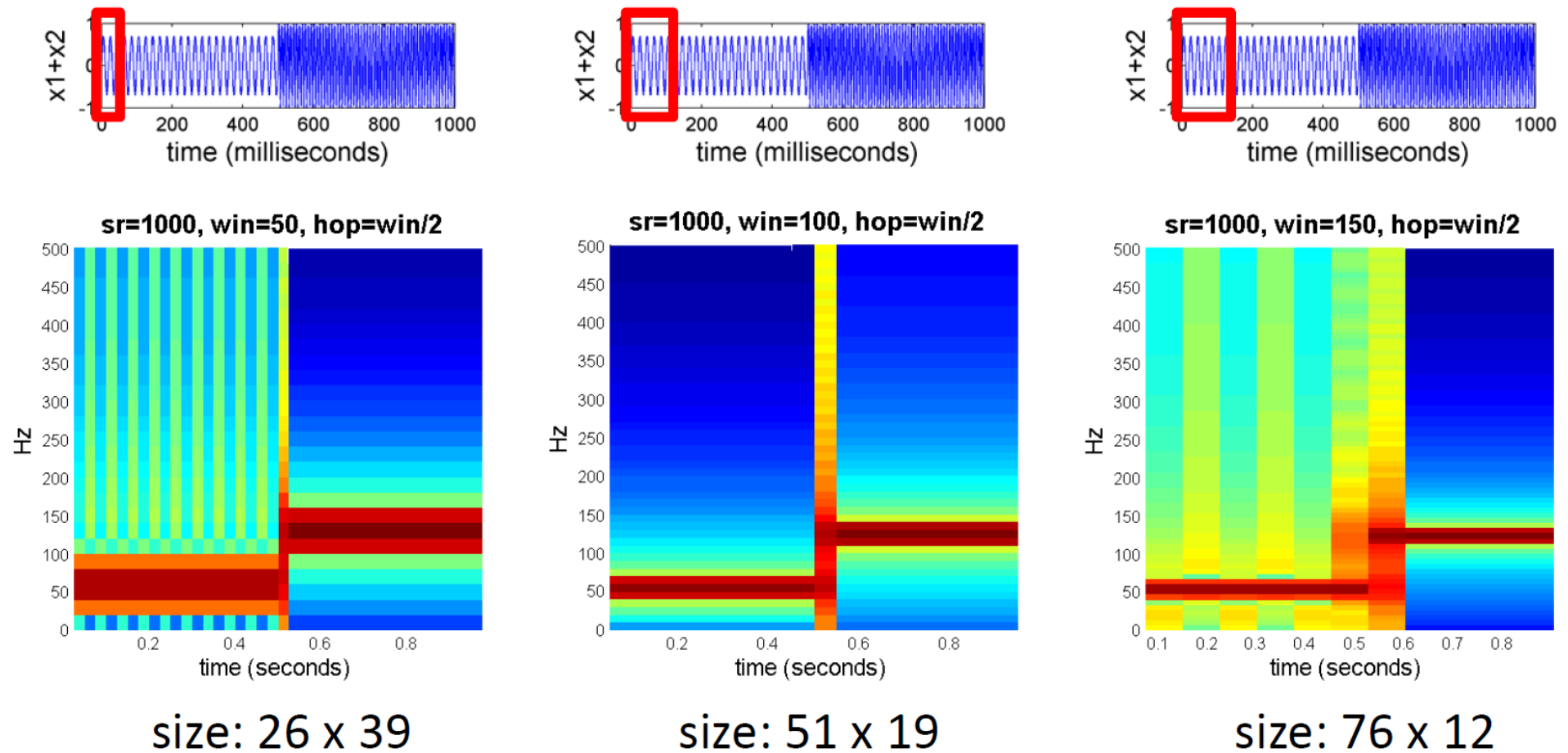
vocal (clean)



Understanding STFT

(assuming the hop size is proportional to the window size)

- Different window size (win_size = 50, 100, 150)

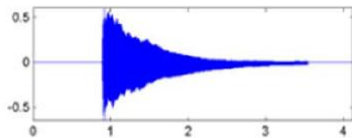


- Shorter window $\rightarrow$ worse **frequency resolution**
- Longer window $\rightarrow$ worse **temporal resolution**

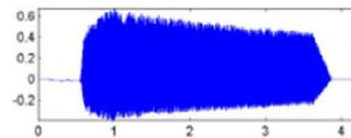
Spectral Characteristics

- Fundamental frequency (**F0**)
- **Partials** (harmonics): usually at frequencies which are integer multiples of F0
 - The partials in different instruments may exhibit **relative strengths**
 - The frequency may deviate from the ideal harmonics (this is called “inharmonicities”)

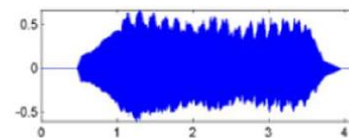
Piano



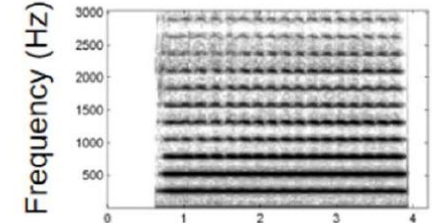
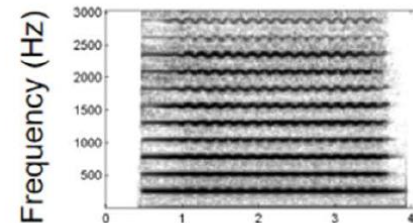
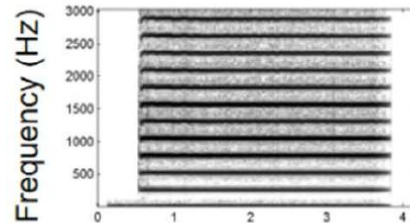
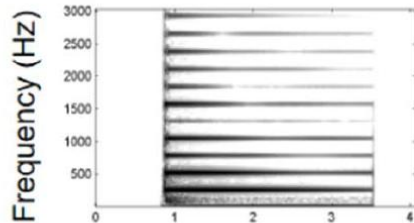
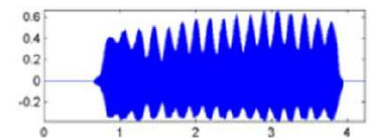
Trumpet



Violin



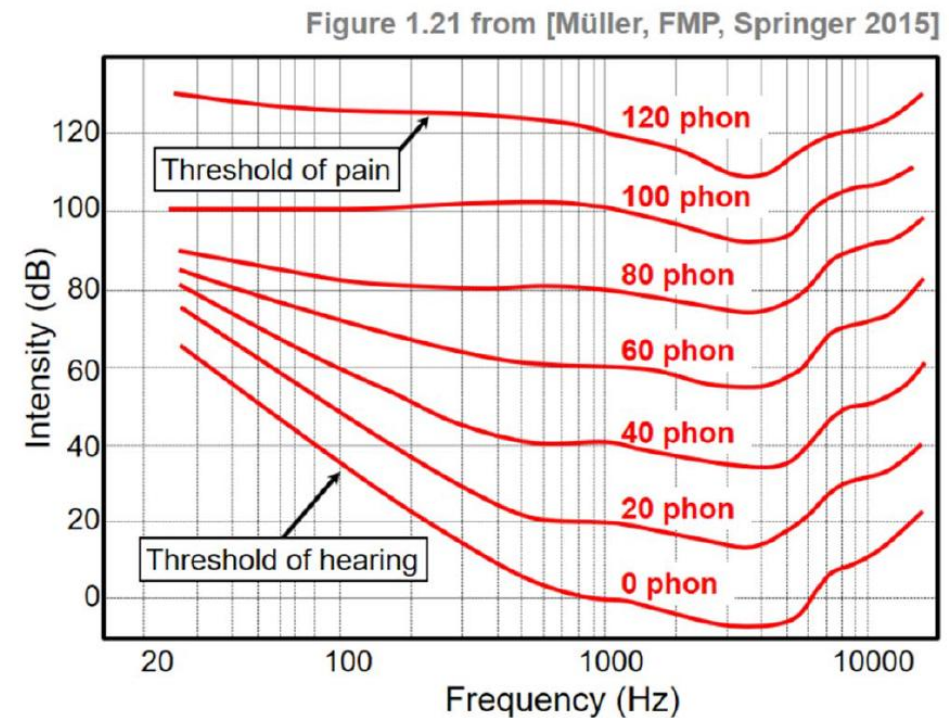
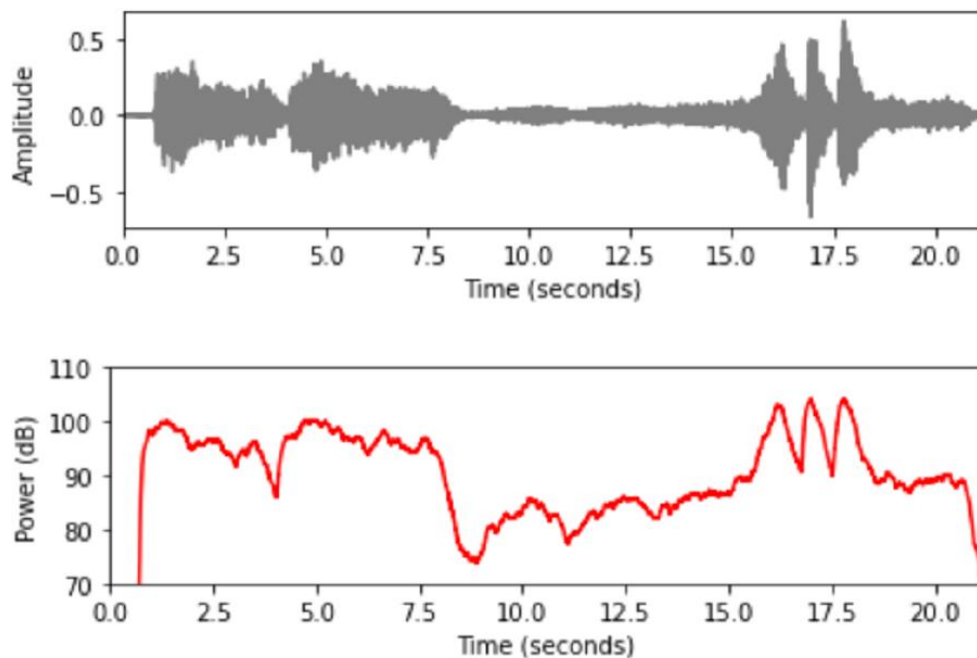
Flute



Dynamics, Intensity, and Loudness

https://www.audiolabs-erlangen.de/resources/MIR/FMP/C1/C1S3_Dynamics.html

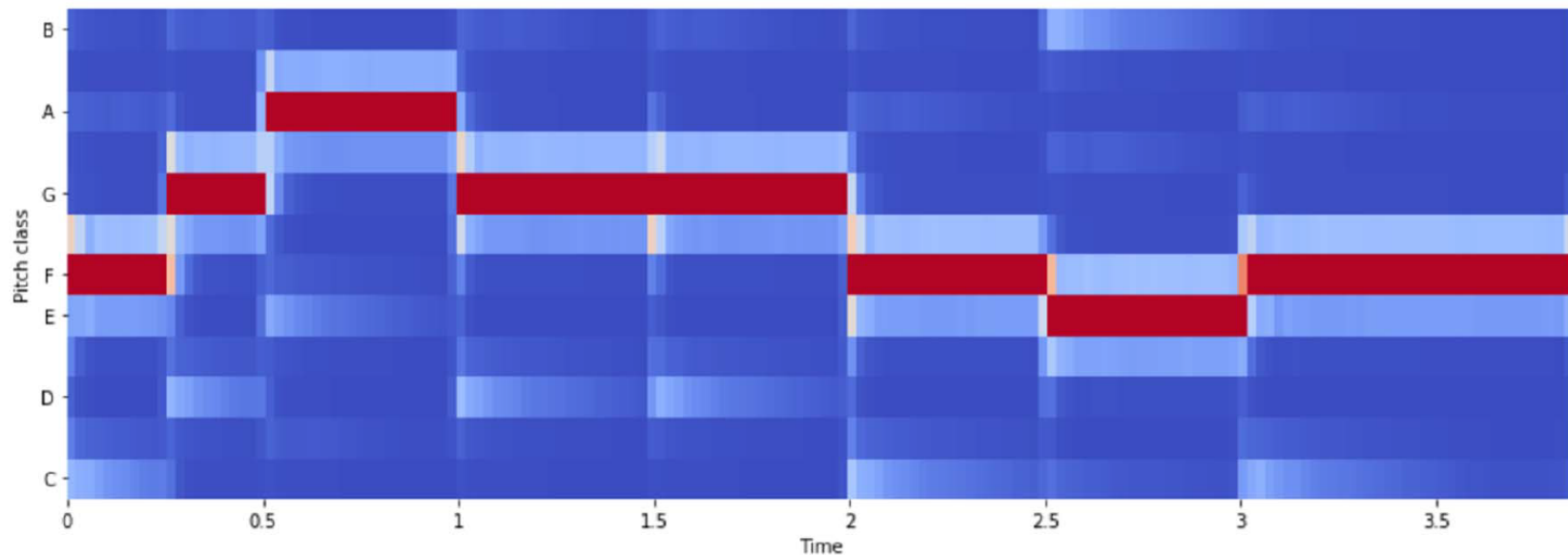
- **Intensity:** a physical property
 - Other names: power (dB); energy
- **Loudness:** a perceptual property
 - Less used in deep learning research



Pitch Class Profile / Chromagram

<https://musicinformationretrieval.com/chroma.html>

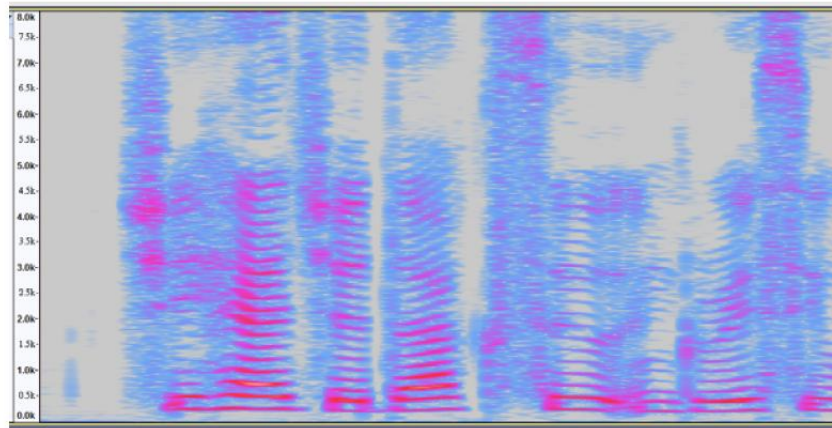
- “A **chroma vector** ([Wikipedia](#)) is a typically a 12-element feature vector indicating how much energy of each pitch class, {C, C#, D, D#, E, ..., B}, is present in the signal”
 - i.e., ignore octaves



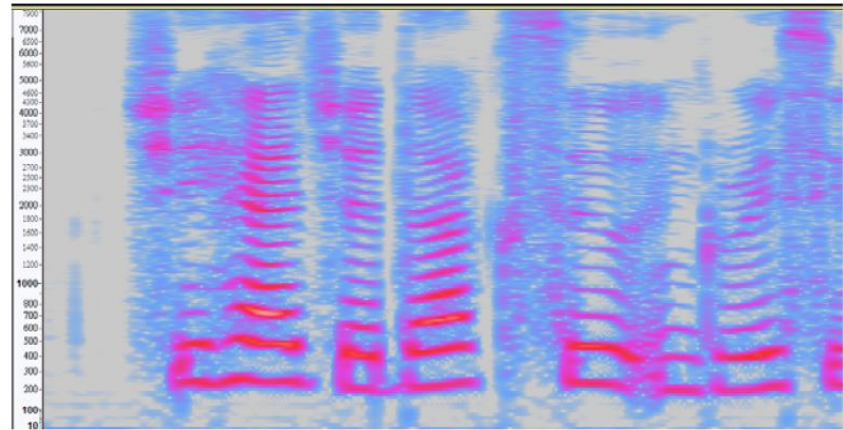
Mel-Spectrogram

- The Mel scale is a perceptual scale of pitches judged by listeners to be equal in distance from one another
- **Finer** resolution in the **low-frequency** range (NOT exactly logarithmic scale)
- Dimension reduction

linear scale



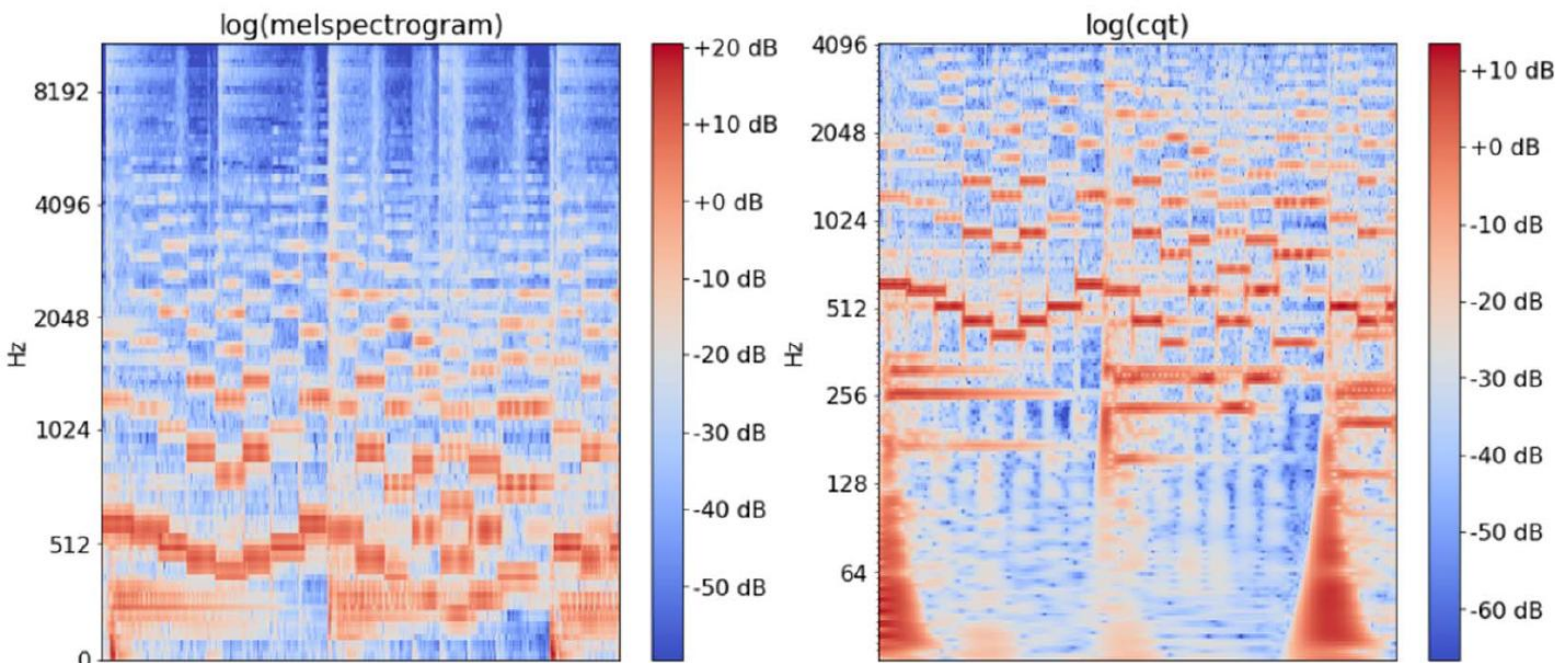
mel scale



Constant-Q Transform (CQT)

https://music-classification.github.io/tutorial/part2_basics/input-representations.html

- **STFT** (*linearly*-spaced frequencies)
- **CQT** (*logarithmically*-spaced, closer to human auditory perception)



– **Logarithmic frequency, logarithmic magnitude**

Feature Extraction

- **Timbre** representation: Spectrogram $\rightarrow$ mel-spectrogram $\rightarrow$ MFCC
- **Harmonic** representation: Spectrogram $\rightarrow$ CQT $\rightarrow$ chroma feature

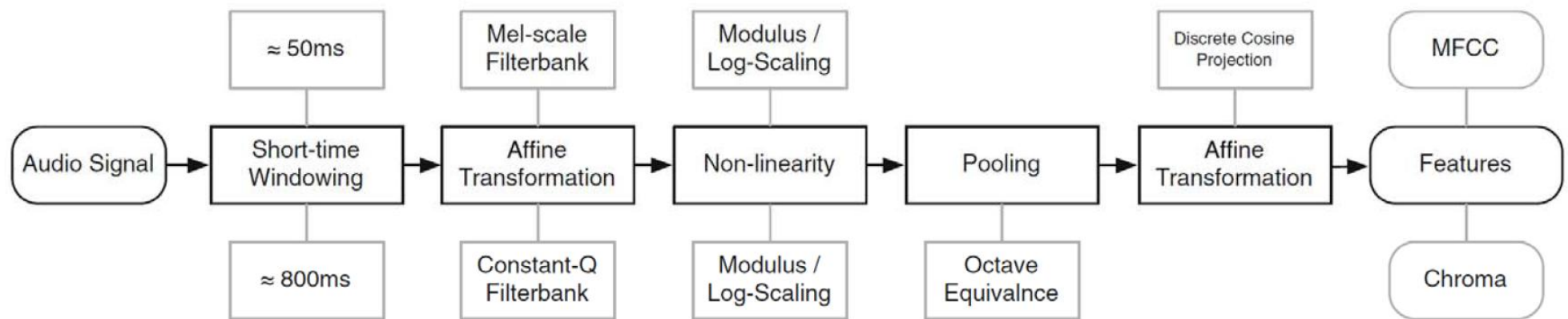


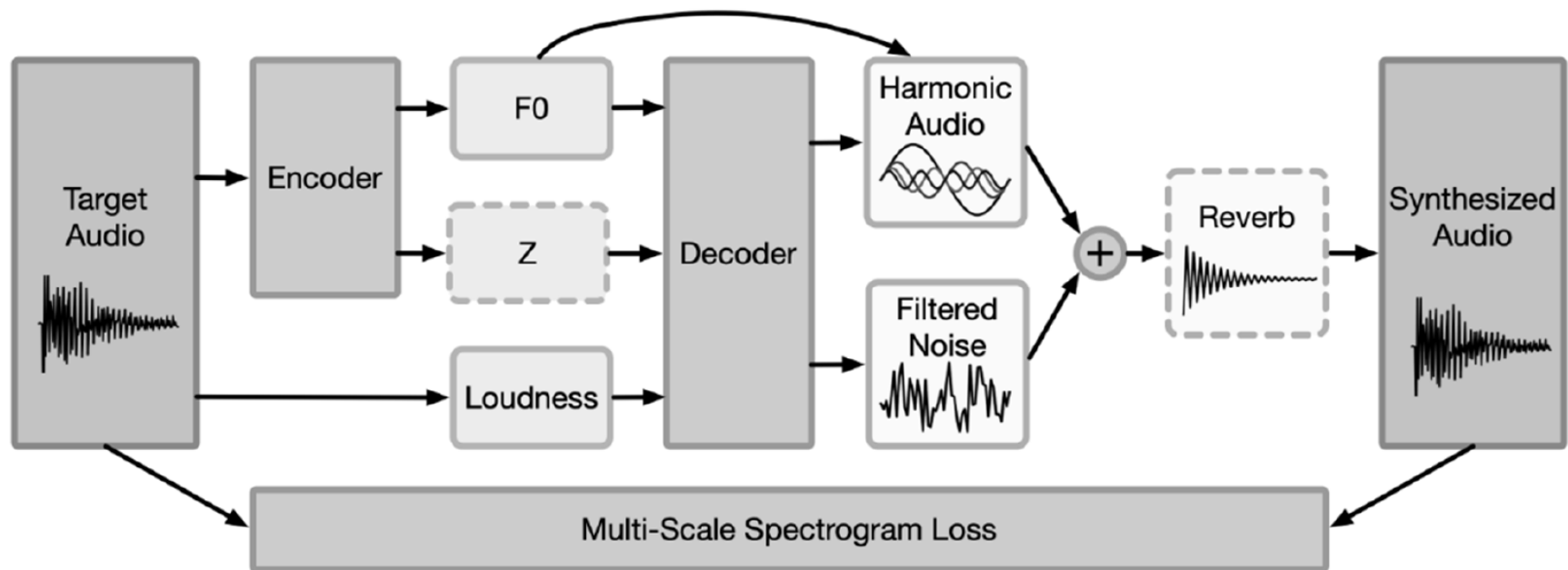
Fig. 3 *State of the art*: standard approaches to feature extraction proceed as the cascaded combination of a few simpler operations; on closer inspection, the main difference between chroma and MFCCs is the parameters used

Decomposition of Elements of Music

(for monophonic signals)

<https://magenta.tensorflow.org/ddsp>

- **Music = F0 (pitch + duration) + loudness + “z” (timbre)**
content *style*



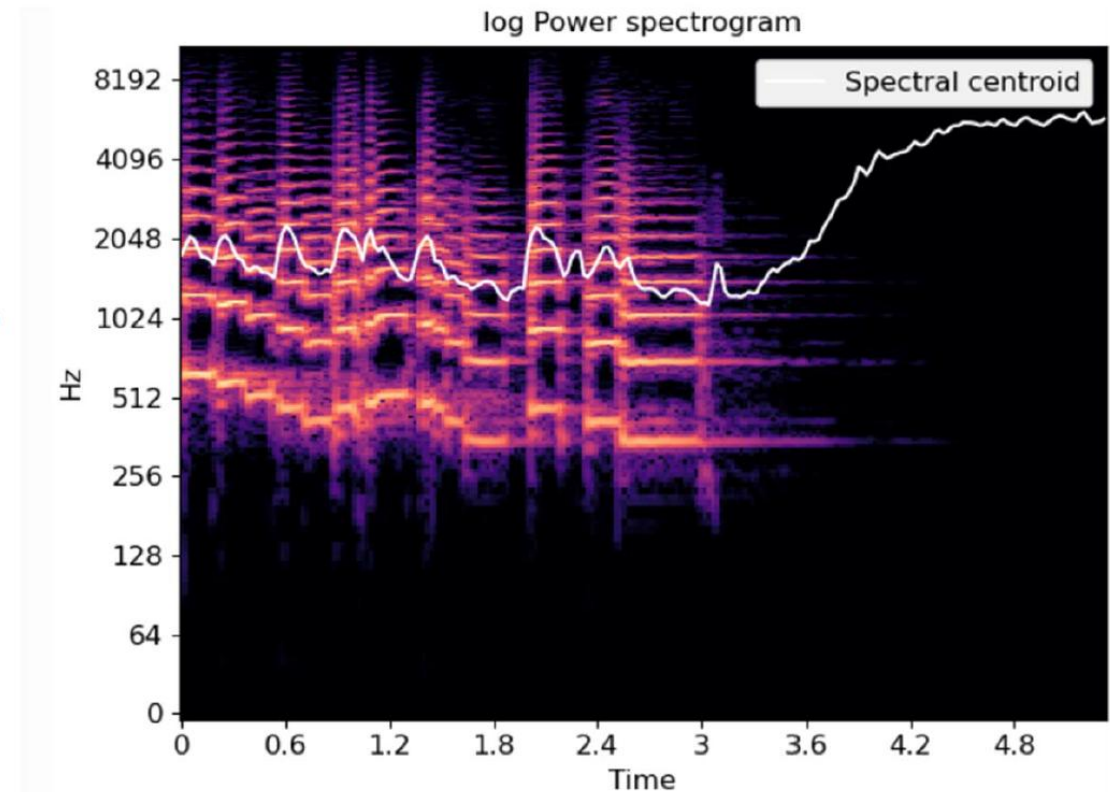
Ref: Engel et al., “DDSP: Differentiable digital signal processing,” ICLR 2020

Audio classification

- Audio classification has a long history originating from speech recognition
 - ▣ Classify audio signals into music, speech, and environmental sounds
 - ▣ Classify musical instrument sounds and sound effects
- The features they used are not adequate for automatic musical genre classification

Spectral Centroid

- Each frame of a magnitude spectrogram is normalized and treated as a distribution over frequency bins, from which the mean (centroid) is extracted per frame
- A measure of spectral shape
- Higher centroid values correspond to “**brighter**” textures with more high frequencies
- Other statistics can also be used
 - bandwidth, skewness, kurtosis



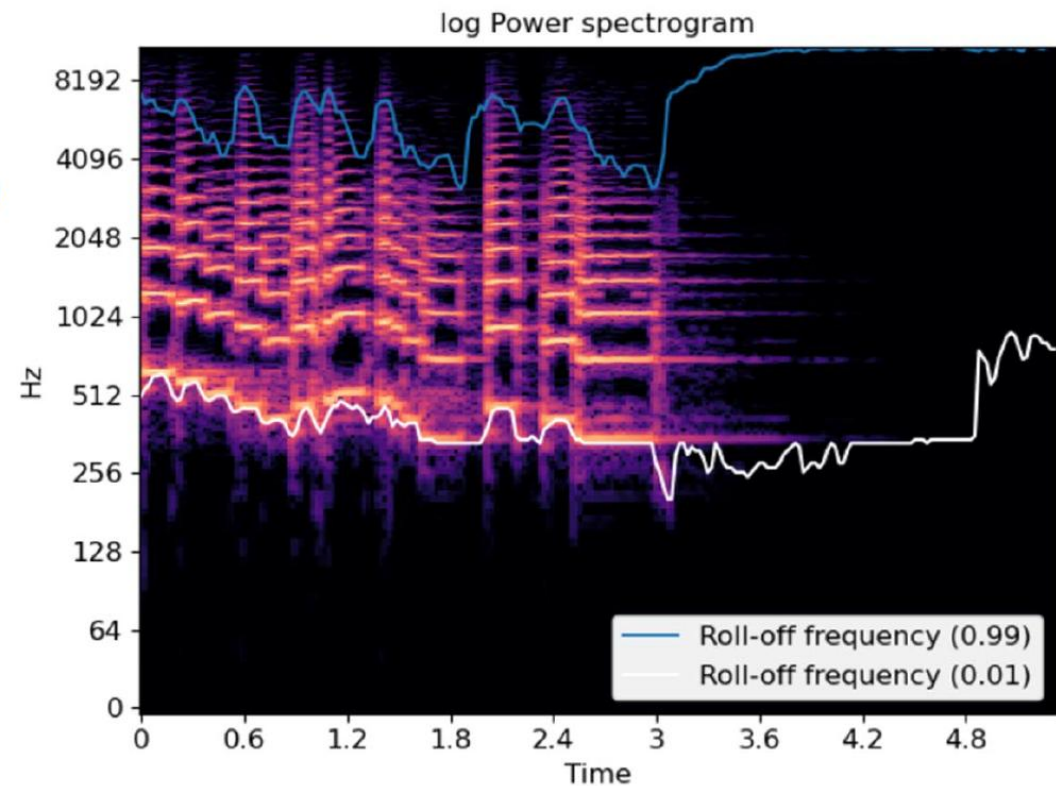
Ref: Tzanetakis and Cook, “Musical genre classification of audio signals,” TASLP 2002

27

頻譜中心代表聲音的“質心”，其數值越小，代表越多的頻譜能量集中在低頻範圍內

Spectral Rolloff

- The frequency for a spectrogram bin such that at least *roll_percent* (0.85 by default) of the **energy** of the spectrum in a frame is contained **in this bin and the bins below**
- Can be used to approximate the **maximum** (or **minimum**) frequency by setting *roll_percent* to a value close to 1 (or 0)
- Another measure of spectral shape



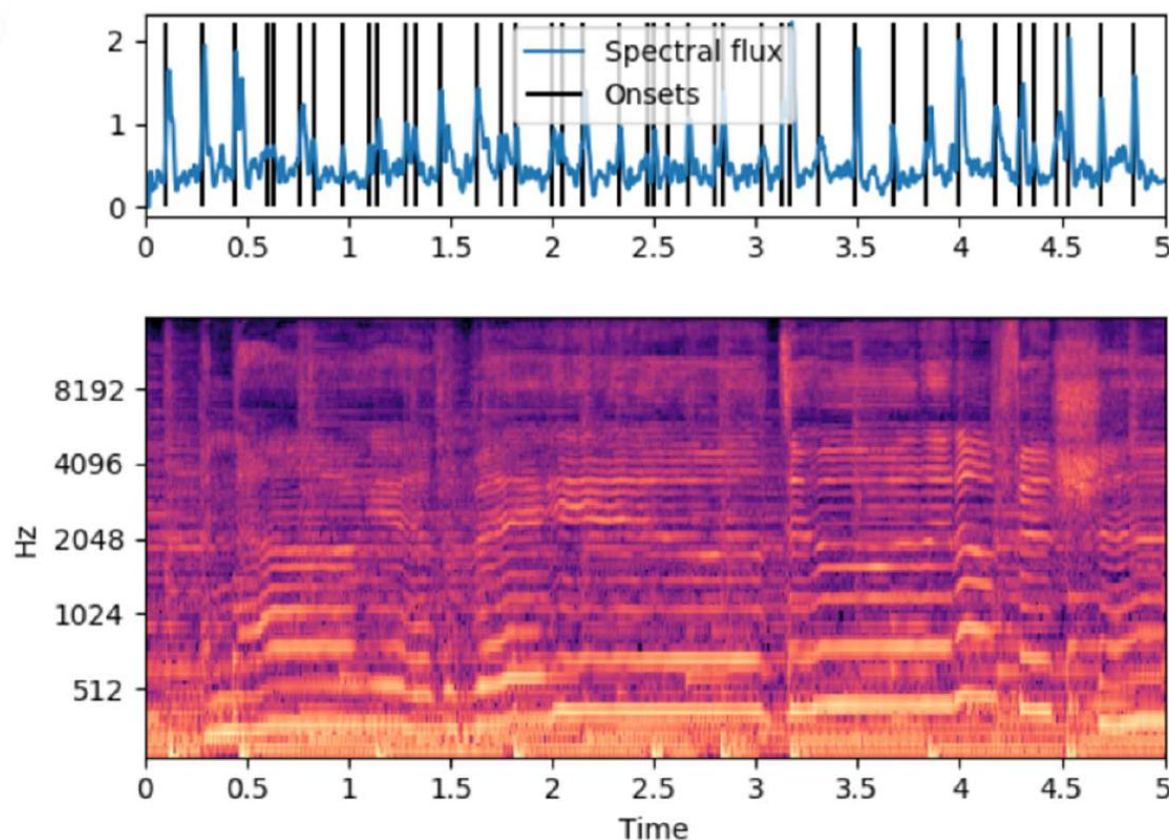
Ref: McFee et al., "librosa: Audio and music signal analysis in python," 2015

29

比該頻率低的頻率的所有能量大於一定比例的整個頻譜的能量，通常這個比例為0.85。

Spectral Flux

- How quickly the power spectrum of a signal is changing over time
- Usually calculated as the L2-norm between two adjacent normalised spectra
- Usually used for musical onset detection



Ref1: https://en.wikipedia.org/wiki/Spectral_flux

Ref2: https://librosa.org/librosa_gallery/auto_examples/plot_superflux.html

Zero crossing rate

- A measure of the noisiness of the signal

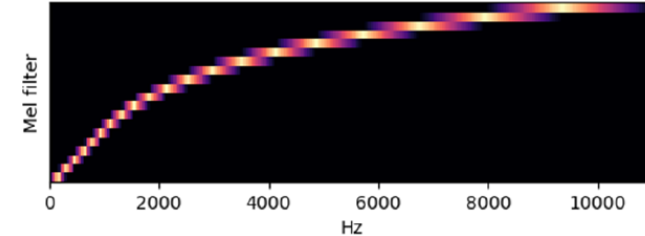
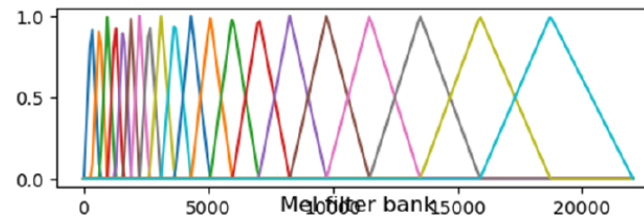
$$Z_t = \frac{1}{2} \sum_{n=1}^N |sign(x[n]) - sign(x[n-1])|$$

sign function is 1 for positive arguments and 0 for negative arguments
 $x[n]$ is the time domain signal for frame t

- Unvoiced speech has a low volume but a high ZCR

ZCR是指在每幀(frame)資料中, 信號通過零點(正變成負, 或負變成正)的次數, 此特徵在語音辨識和音訊信息檢索領域廣泛應用, 是金屬聲音和搖滾樂的關鍵特徵

Mel-frequency cepstral coefficients (MFCC)



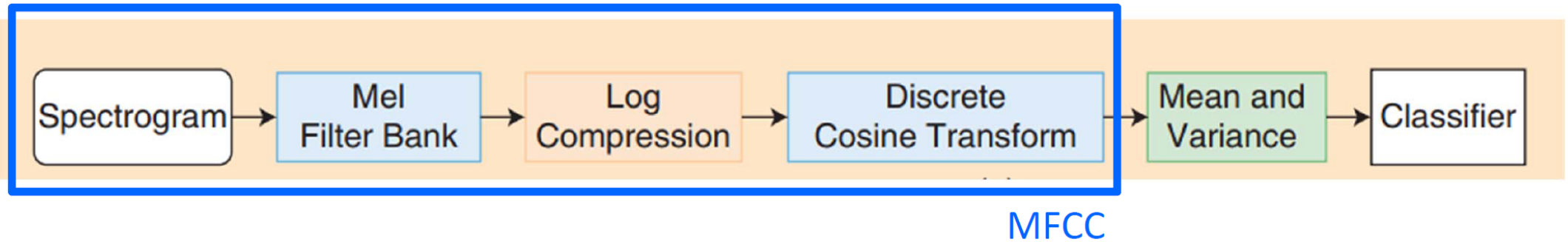
- Procedure
 1. Log-amplitude of the magnitude spectrum
 2. Grouping the FFT bins according to the *perceptually motivated* Mel-filter bank
 - **Linear** till 1,000 Hz and **logarithmic** above it
 - Usually with triangular overlapping windows
 3. Taking logs and DCT for uncorrelating the resulting features
- **Compact representation** of the spectrum (1,024-dim $\rightarrow$ 128 or 13 coefficients)
 - Somehow capture the relative strength of partials (without knowing the F0)
 - Less interpretable

Ref: McFee et al., “librosa: Audio and music signal analysis in python,” 2015

31

人類聽覺天生對不同頻率有不同的反應, 簡單以頻率變化敏感度來說, 低頻>高頻但響度(聽覺上的大小聲)的敏感度卻反過來是高頻>低頻

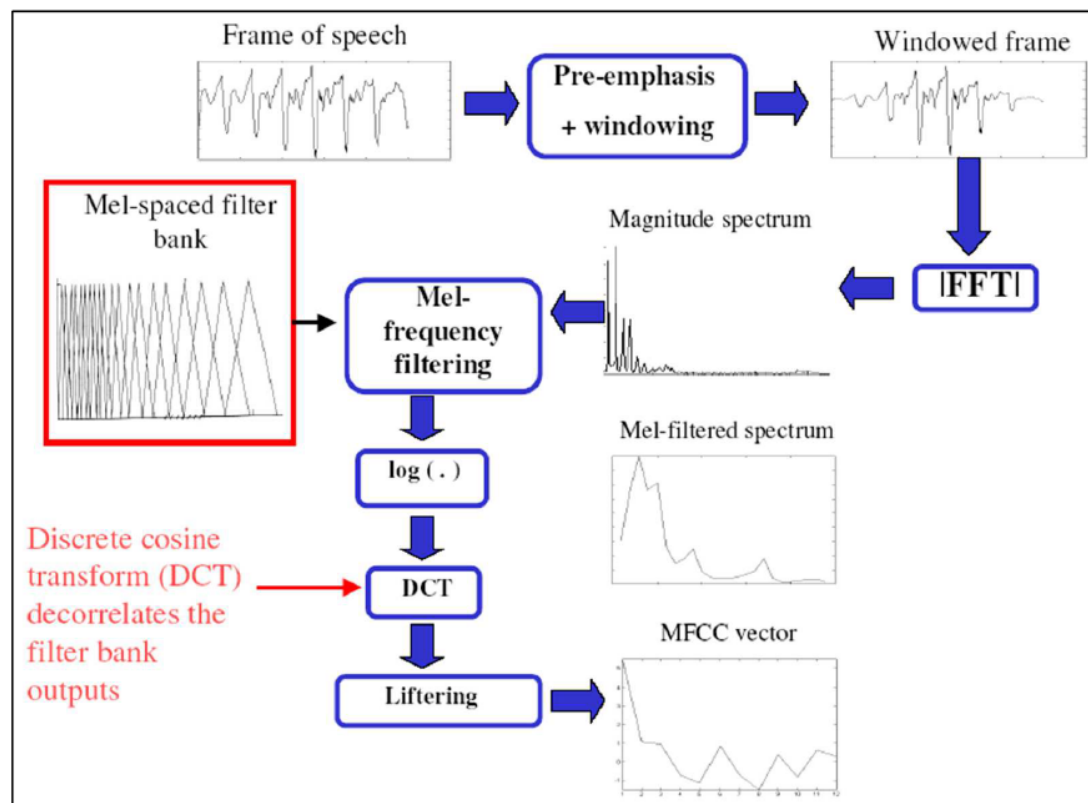
Spectral Features Can be Used to Build a Classifier



- Procedure
 1. Compute the features per STFT frame (e.g., 13-D frame-level features)
 2. **Temporal pooling** over time for each audio clip (e.g., by taking the **mean and variance**; leading to 26-D clip-level features)
 3. Use that as input to a **classifier** (e.g., random forest, or support vector machine)
- Limits
 - *Clear* physical meaning but not sophisticated enough and *limited semantic* meaning
 - Only the classifier is trainable

Mel-Frequency Cepstral Coefficients (MFCC)

- First five coefficients provide the best genre classification performance



$$X_a[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi nk/N}, \quad 0 \leq k < N$$

$$S[m] = \ln \left[\sum_{k=0}^{N-1} |X_a[k]|^2 H_m[k] \right], \quad 0 < m \leq M$$

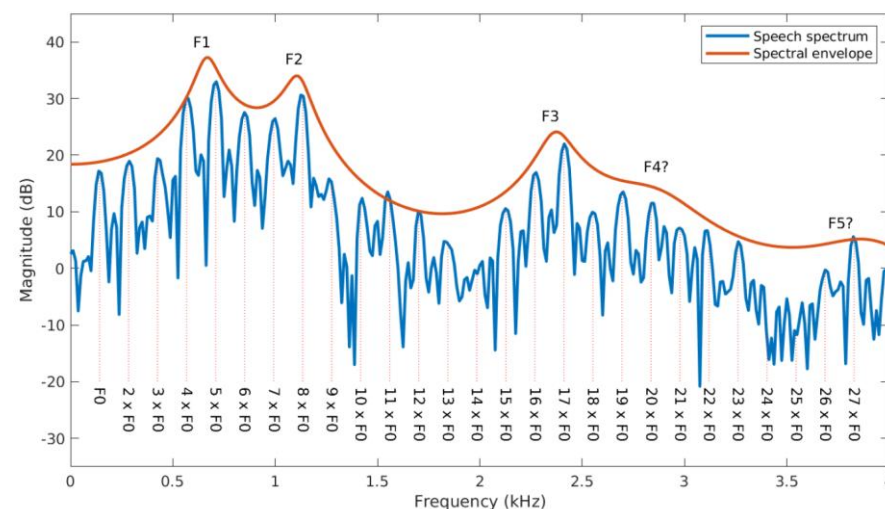
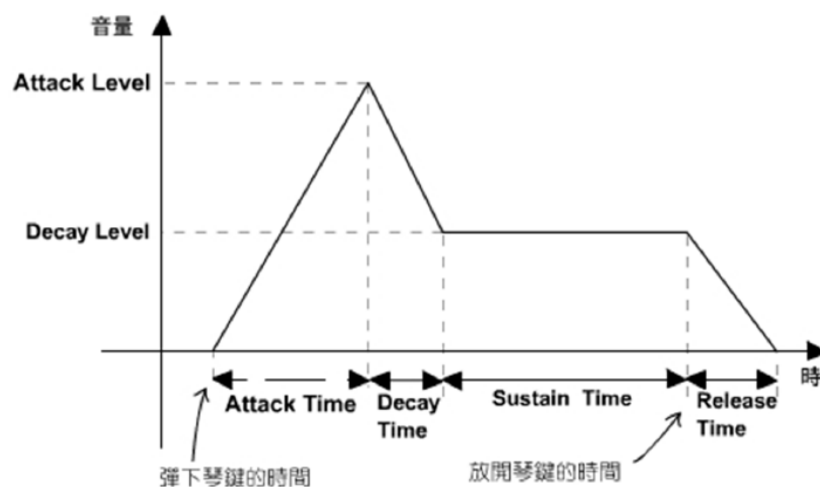
$$c[n] = \sum_{m=0}^{M-1} S[m] \cos(\pi n(m-1/2)/M), \quad 0 \leq n < M$$

M : the number of filters

N : the size of the FFT

Envelope

- 將一種音色波形的大致輪廓描繪出來，就可以表示出該音色在音量變化上的特性，而這個輪廓就稱為Envelope(波封)
- 一個波封可以用4種參數來描述，分別是 Attack(起音)、Decay(衰減)、Sustain(延持)、與 Release(釋音)，這四者也就是一般稱的"ADSR"。



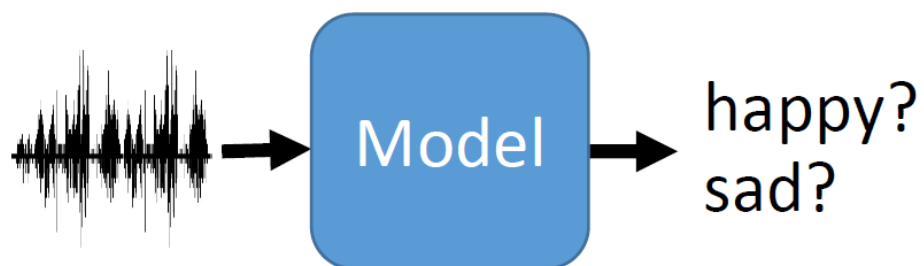
Visual Information

Input video (two people speaking together)

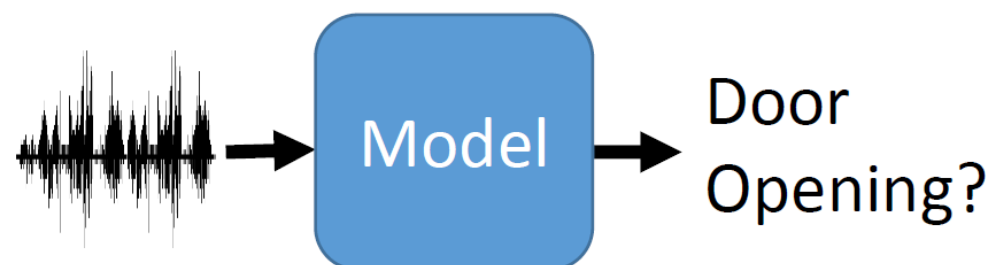


Video source: Team Coco, <https://www.youtube.com/watch?v=UT7h4nRcWjU>

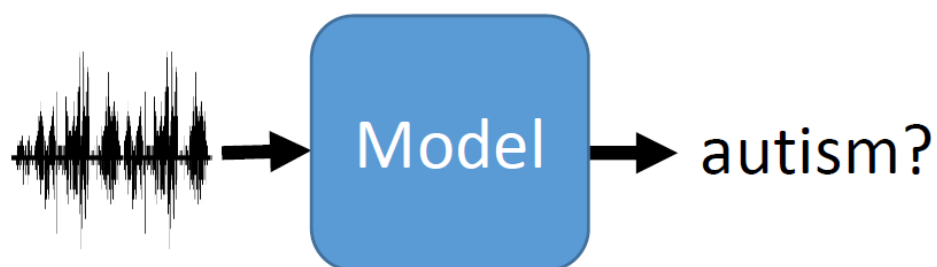
Speech applications



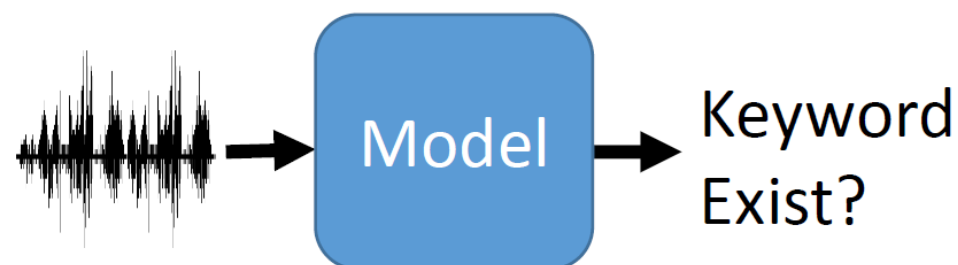
Emotion Recognition



Sound Event Detection



Autism Recognition

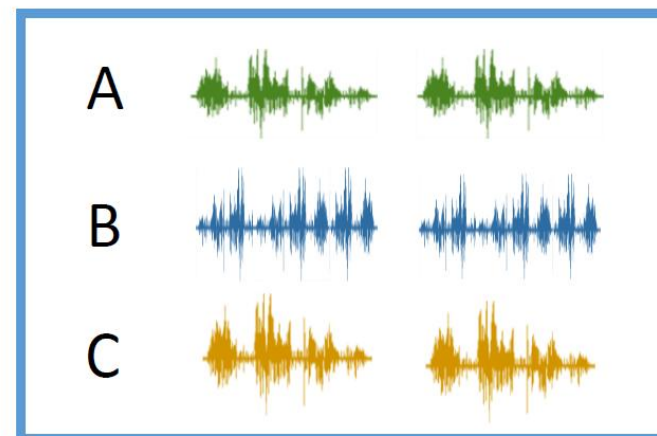


Keyword Spotting

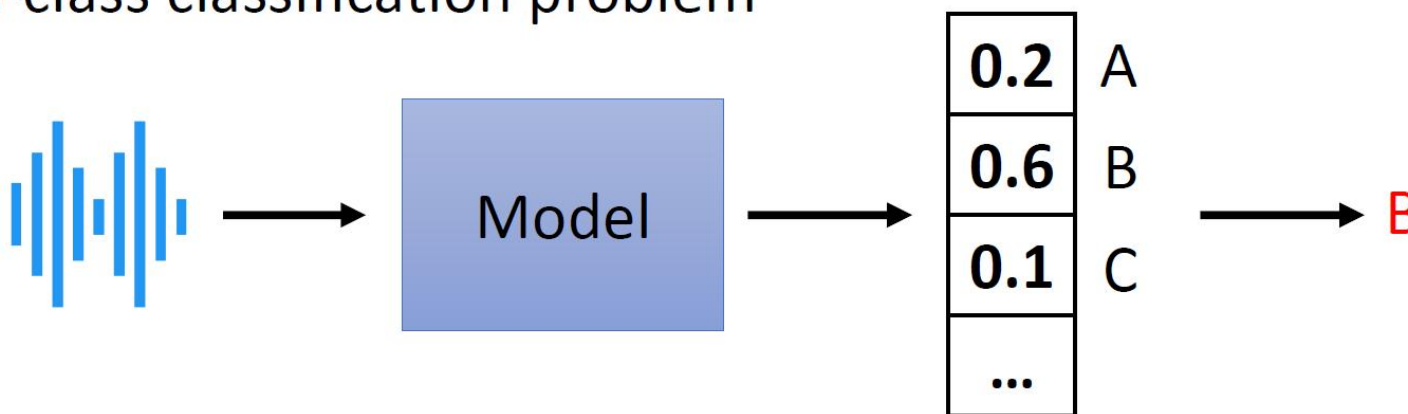
Speech applications

- Speaker Recognition / Identification

- 語者識別
- 一段語音是誰所說的



A multi-class classification problem



Speaker Verification

